

CSC 1102 English Composition & Comprehension

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: English Composition & Comprehension	Semester: 1 Credits: 3
Course Code: CSC 1102	Course Type: Core
Course Area/Domain: GER	Contact Hours: 3 hours/Week
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

COURSE OBJECTIVES

The core objective of this course is to enable students to become better listeners, readers and writers. This includes the ability to use appropriate words according to the context, construct coherent paragraphs through summarizing, paraphrasing, process analysis, cause and effect, comparison and contrast and making oral presentations effective through an understanding of non- verbal means of communication.

Course Description

This comprehensive English language course aims to equip students with essential skills for effective communication, focusing on enhancing their abilities as listeners, readers, and writers. Through a structured curriculum covering topics such as paragraph and essay writing, summarizing, paraphrasing, expository and argumentative writing, data analysis, punctuation, reading comprehension, and vocabulary building, students will develop a strong foundation in using language appropriately, constructing coherent paragraphs and essays, and delivering engaging oral presentations.

Text/Reference Books

Real Writing with Readings by Susan Anker (Sixth Edition)
7 habits of highly effective teenagers, Sean Covey, 1st Edition 2006.

SYLLABUS

Week	Topics
1	Introduction Session, Warming-up, Course Outline Orientation
2	Reading between the lines Reading for Meaning and Understanding Reading for Details.
3	Paragraph Writing/Topic Sentence/Supporting Evidence/Concluding Sentence/ Cohesion/Coherence.
4	Structuring an essay/Thesis Statement and Main Ideas/Paragraphing and Cohesion/ Describing a place, person, thing, idea, etc./Applying Descriptive Writing Strategies.
5	Narrative Writing/Developing Picture Stories/Writing accounts and stories/Making Mind Map/ Characters/ Rising Action/ Climax/Falling Action/Personification/ Imagery/ Flashback
6	Summarizing/Reading for Meaning/Reading for Ideas/Identifying Key Ideas/Avoiding Block lifting
7	Paraphrasing: Rewriting to eliminate plagiarism/Reading for Main Idea/Identifying Key Ideas/Changing the Order of Information/ Removing Redundancy/Giving Citation
8	Midterm
9	Expository writing, looking at a discussion topic in a balanced way, essay writing.
10	Argumentative Writing, arguing for both sides while taking a clear stance.
11	Studying and analyzing data, Putting it in shape using appropriate language and structure.
12	Punctuation and Parts of Speech, Discussion on punctuation marks i.e. Full Stop/Comma/Exclamation, Mark/Question, Mark/Colon/Semicolon/Speech Marks/Parentheses Reinforcement of all parts of speech.
13	Reading Comprehension Skimming/Scanning/ Inferring/ Evaluating Deduction and Induction.
14	Vocabulary Building Synonyms and Antonyms Homonyms and Homophones.
15	Listening Comprehension Listening for specific details, Listening for Gist.

COURSE LEARNING OUTCOMES (CLOs):

CLO	Description	Domain	BT Level*
1	Demonstrate comprehension and critical analysis of written texts, including skimming, scanning, and identifying key ideas.	C	2
2	Apply effective writing strategies in paragraphs, essays, and various compositions with proper structure, cohesion, and coherence.	C	3
3	Utilize paraphrasing, summarizing, and proper punctuation to communicate ideas clearly and avoid plagiarism	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1				x						
CLO.2			x							
CLO.3							x			

GRADUATE ATTRIBUTES (GAs):

GA	Description
GA-3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
GA-4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
GA-7	Communicate effectively with the computing community about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

CSC 1103 Fundamentals of Programming

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Fundamentals of Programming	Semester: 1 Credits: 3
Course Code: CSC 1103	Course Type: Core
Course area/domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): CSCL 1103	Lab Course Name: Lab: Fundamentals of Programming
Course Pre-Requisite: None	

Course Objective

The objectives of this course are to: 1. Introduce the fundamental concepts of programming and problem-solving. 2. Familiarize students with the syntax, semantics, and structure of the C programming language. 3. Develop the ability to analyze problems and design algorithms for their solutions. 4. Provide hands-on experience in using arrays, strings, pointers, and structures for data handling. Introduce basic concepts of file handling for input/output operations.
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Course Description

This course introduces the fundamental concepts of computer programming using the C language. It is designed for beginners with little or no prior programming experience. The course covers problem-solving techniques, algorithm development, and structured programming concepts. Students will learn the syntax and semantics of C, including variables, data types, operators, control structures, functions, arrays, pointers, strings, and file handling. Emphasis is placed on writing efficient, well-structured, and error-free code through hands-on practice and programming assignments. By the end of the course, students will be able to design, implement, debug, and document C programs to solve basic computational problems, providing a solid foundation for advanced courses in programming and computer science.

Text Book

C How to Program, Paul Deitel & Harvey Deitel, 7th Edition

Reference Book

The C Programming Language, Brian W. Kernighan, Dennis M. Ritchie, 2nd Edition

SYLLABUS

Week	Topics
1	Introduction to Programming
2	Basics of C Language, Data types, variables, constants
3	Operators & Expressions
4	Control Structures – Decision Making
5	Control Structures – Loops
6	Problem-Solving with Loops
7	Functions
8	Mid Term
9	Arrays
10	Strings in C
11	Pointers in C
12	Dynamic Memory Allocation
13	Structures & User-Defined Data Type
14	File Handling
15	Review & Project Discussion

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Understand basic programming concepts, operators, and control structures to implement C programs.	C	2
2	Develop modular programs using functions, arrays, and structures to solve computational problems	C	3,6
3	Implement file handling, pointers, and dynamic memory allocation to design efficient C programs.	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

GRADUATE ATTRIBUTES (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	x									
CLO.2				x						
CLO.3					x					

GRADUATE ATTRIBUTES (GAs)

GA	Description
GA-1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
GA-4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
GA-5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

CSCL 1103 Lab Fundamentals of Programming

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Fundamentals of Programming	Semester: 1 Credits: 1
Course Code: CSCL 1103	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code (If Any): CSC 1103	Theory Course Name: Fundamentals of Programming
Course Pre-Requisite: None	

Course Description

This course covers the fundamental concepts of programming using C language. Student will be equipped with all important concepts and techniques required to be a successful programmer.	
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Course Objectives

The objectives of this course are to: Learn about problem solving skills, writing pseudo code, getting familiarized with basic building blocks and programming constructs of a fundamental programming language and enabling Students to develop analytical skills and solve practical programming problems	
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TEXT/REFERENCE BOOKS:

C Programming Absolute Beginners Guide, Greg Perry, Dean Miller, 3rd Edition, Que Publishing, 2013.

SYLLABUS

Week	Topics
1	Programming Languages, Introducing C
2	C Fundamentals
3	Formatted Input/Output
4	Expressions, Basic Types
5	Selection Statements
6	Loops, Nested Loops
7	Arrays
8	Midterm
9	Functions
10	Program Organization, Debugging, Error Tracking
11	Pointers, Pointers for Array Processing
12	String, [Searching, Sorting
13	Structures, Unions, Enumerations
14	Filing
15	Project Evaluation and Demonstration

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Apply fundamental C programming concepts to solve computational problems.	C	3
2	Analyze and debug C programs using modular design and error-tracking techniques.	C	4
3	Use pointers, strings, structures, unions, enums, and file handling to develop structured programs.	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs):

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		x								
CLO.2			x							
CLO.3				x						

GRADUATE ATTRIBUTES (GAs):

GA	Description
GA-2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
GA-3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
GA-4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

CSC 1108 Introduction to Computer Science

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Introduction to Computer Science	Semester: 1 Credits: 2
Course Code: CSC 1108	Course Type: Core
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding lab Course code (if any): CSC 1108	Lab Course Name: Lab: Introduction to Computer Science
Course Pre-Requisite: None	

Course Description

This course provides an introduction to the fundamental concepts of computing and information technology. It covers the history and evolution of computers, computer hardware and software, operating systems, data representation, and the role of computing in society. Students will also gain hands-on experience with productivity tools, basic problem-solving techniques, and an introduction to programming.

Course Objectives

The main objectives of this course are to:

1. Introduce the basic concepts and terminology of computing and information technology.
2. Provide an understanding of computer hardware, software, and operating systems.
3. Develop problem-solving and logical thinking skills using basic programming concepts.
4. Equip students with practical skills in productivity software (e.g., word processing, spreadsheets, presentations).

Familiarize students with the web development and databases

TEXT/REFERENCE BOOKS:

Introduction to Computers, Peter Norton

Introduction to Computers, G. M. Long & D. Morley

Think Python: How to Think Like a Computer Scientist, Allen B. Downey

SYLLABUS

Week	Topics
1	Introduction to Computers, Hardwares
2	Softwares, types of softwares
3	Number Conversions
4	Introduction to HTML
5	More about HTML and CSS
6	Introduction to DBMS
7	Queries in MySQL
8	Mid Term
9	Logic Building and Algorithm Designing
10	Introduction to Python, Basics of Python
11	Variables, basic operators, decision making in python
12	Loops, nested loops in python
13	Strings, arrays and lists in python
14	Introduction to AI
15	AI using Python

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Understand computer fundamentals, hardware/software, number systems, and web development basics (HTML/CSS).	C	2
2	Apply algorithm design, Python programming, and decision-making constructs to solve computational problems.	C	3
3	Implement database queries and basic AI applications using Python to develop simple computing solutions.	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	x									
CLO.2				x						
CLO.3					x					

GRADUATE ATTRIBUTES (GAs)

GA	Description
GA-1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
GA-4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
GA-5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

CSCL 1108 Lab Introduction to Computer Science

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Introduction to Computer Science	Semester: 1 Credits: 1
Course Code: CSCL 1108	Course Type: Core
Course Area/Domain: GER	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code(If any): CSC 1108	Course Name: Introduction to Computer Science
Course Pre-Requisite: None	

Course Description

This course covers introduction to computers and modern digital tools. It covers essential office applications, basics of web development, foundational programming concepts, version control, and introductory database management. Emphasis is placed on hands-on practice to build problem-solving and technical skills.	This course covers topics covered in Computers; The Devices, Mac Microcomputer Software; Intro Networks; Data Collaborative Security and o
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Course Objectives

Main objective of this course is to:

1. Introduce students to the field of computing, computing environment, various types of software and hardware.
 2. The course also involves the introduction to the subfields of computer science.
- Makes students able to learn all the basics about different areas of computer science including database systems, networking, artificial intelligence and programming.

TEXT/REFERENCE BOOKS

Introduction to Computers, Peter, 6th Edition, McGraw-Hill.
Fundamentals of Information Technology

SYLLABUS

Week	Topics
1	MS Word – Document creation, formatting, tables
2	MS Excel – Formulas, functions, sorting/filtering, charts, Pivot Table intro
3	MS PowerPoint – Slide design, animations, professional presentation
4	Web Development (HTML,CSS)
5	Web Development (HTML,CSS)
6	Introduction to DBMS
7	DBMS
8	Mid-Term
9	Python Fundamentals
10	Python Loops
11	Python for AI
12	Python for AI
13	Git & Version Control – Git setup, init repo, commit, push/pull
14	Capstone Project Work
15	Final Lab

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Use MS Word, Excel, and PowerPoint to produce professional documents, spreadsheets, and presentations.	P	3
2	Develop basic webpages using HTML/CSS and perform simple DBMS operations.	C	6
3	Apply Python programming concepts to solve problems, including introductory AI tasks.	C	3
4	Use Git for version control and manage project workflows.	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1					x					
CLO.2		x								
CLO.3			x							
CLO.4						x				

GRADUATE ATTRIBUTES (GAs):

GA	Description
GA-2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
GA-3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
GA-5	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
GA-6	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.

CSC 1107 Applied Physics

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Applied Physics	Semester: 1 Credits: 2
Course Code: CSC 1107	Course Type: Core
Course Area/Domain: GER	Contact hours: 2 hours/Week.
Corresponding lab Course code (If any): CSCL 1107	Lab Course Name: Lab: Applied Physics
Course Pre-Requisite: None	

Course Description

This course introduces students to the fundamental principles of electromagnetism and optics with a focus on applications in computing and electrical systems. It covers electric and magnetic fields, Gauss's and Faraday's laws, electromagnetic induction, and the behavior of light through reflection, refraction, diffraction, and polarization. Emphasis is placed on problem-solving and conceptual understanding to prepare students for advanced courses in electronics, communication, and computing hardware.

Course Objectives

By the end of this course, students will be able to:

1. Understand the basic concepts of electricity, magnetism, and electromagnetism and their applications.
2. Apply fundamental physical laws such as Gauss's and Faraday's laws to analyze electric and magnetic fields.
3. Comprehend the principles of wave optics including interference, diffraction, and polarization.

Relate physical principles to practical scenarios in computing and electronic systems.

TEXT/REFERENCE BOOKS

Fundamentals of Physics (Extended), 10th edition, Resnick and Walker
 Narciso Garcia, Arthur Damask, Steven Schwarz., "Physics for Computer Science Students", Springer Verlag, 1998.

SYLLABUS

Week	Topics
1	Electric force and its applications and related problems, conservation of charge, charge quantization
2	Electric fields due to point charge and lines of force. Ring of charge, Disk of charge
3	Ring of charge, Disk of charge, A point charge in an electric field, Dipole in a n electric field, The flux of vector field, The flux of electric field
4	Gauss' Law, Application of Gauss' Law, spherically symmetric charge distribution, A charge isolated conductor.
5	Electric potential energy, Electric potentials
6	Calculating the potential from the field and related problem Potential due to point and continuous charge distribution
7	Potential due to dipole, equi-potential surfaces, Calculating the field from the Potential
8	Mid Term
9	Ohm's law and its applications, The Hall effect, The magnetic force on a current
10	The Biot- Savart law, Line of B, Two parallel conductors, Amperes' s Law, Solenoid, Toroids, Electric current, Current density, Resistance, Resistivity and conductivity
11	Faraday's experiments, Faraday's Law of Induction, Lenz's law, Motional emf, Induced electric field, Induced electric fields
12	The basic equation of electromagnetism, Induced Magnetic field, The displacement Current
13	Reflection and Refraction of light waves, Total internal reflection, Two source interference, Double Slit interference, related problems
14	Interference from thin films, Diffraction and the wave theory and their related problems, Single-Slit Diffraction, and related problems
15	Polarization of electromagnetic waves, Polarizing sheets, related problems.

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Explain fundamental concepts of electric, magnetic, and electromagnetic fields and their interactions.	C	2
2	Apply Gauss's Law, Faraday's Law, and related principles to analyze and solve basic electromagnetic problems.	C	3
3	Analyze the phenomena of interference, diffraction, and polarization of light and relate them to real-world applications.	C	4
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND Graduate Attributes (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	x									
CLO.2		x								
CLO.3			x							

GRADUATE ATTRIBUTES (GAs)

GA	Description
GA-1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
GA-2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
GA-3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

CSCL 1107 Lab: Applied Physics

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Applied Physics	Semester: 1 Credits: 1
Course Code: CSC 1107	Course Type: Core
Course Area/Domain: GER	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code (If any): Applied Physics	Course Name: CSC 1107
Course Pre-Requisite: None	

Course Description

This course provides a comprehensive introduction to key concepts in electromagnetism and optics, covering fundamental topics such as charge, current, resistance, and electric fields. Students will explore crucial laws including Coulomb's Law, Gauss' Law, Ampere's Law, and Faraday's Law, which govern the behavior of electric and magnetic fields. Additionally, the course delves into the principles of light, addressing the laws of reflection and refraction, interference, diffraction, and polarization. Through these topics, students will gain a solid foundation in the physical laws essential for advanced studies in physics.

Course Objectives

By the end of this course, students will be able to:

1. Conduct experiments related to electricity, magnetism, and light to reinforce theoretical understanding.
 2. Use basic laboratory instruments such as multimeters, oscilloscopes, and power supplies for data collection.
 3. Analyze experimental data to verify physical laws and interpret measurement uncertainties.
- Work effectively in teams to perform laboratory tasks and communicate results in technical reports.

TEXT/REFERENCE BOOKS

1. Electronic devices, Thomas L. Floyd, 10th Edition, ,2015.
2. Applied Physics, Dale Ewen, Neill Schurter, Erik Gundersen, 11th Edition, Pearson; 11th edition (January 17, 2016),2007.

SYLLABUS

Week	Topics
1	Overview of the Multimeter, Oscilloscope, Trainer Kit, and Associated Software
2	Components of input, output, and storage devices as well as computers
3	Resistor parallel combinations, Testing both on Software and on Hardware
4	Resistor Series & Parallel Combination, Testing both on Software and on Hardware
5	Capacitors series combination, Testing both on Software and Hardware
6	Capacitors parallel combination, Testing both on Software and Hardware
7	Ohm's law verification and implementation both on software and hardware, Ohm's law verification with a variable voltage source, implementation both on software and hardware
8	MID TERM
9	Verification of Faraday law of Electromagnetic Induction through coil and bar magnet experiment
10	Ampere's Law/Lenz's Law Implementation and Verification
11	Exploring the Laws of Reflection: Understanding Light Behavior on Reflective Surfaces
12	Investigating the Refraction of Light: Analyzing Light Behavior Across Different Media
13	Diffraction of Light
14	Step up /down AC voltages
15	Alternative Current to Direct Current Conversion

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Perform and demonstrate experiments related to electricity, magnetism, and optics using appropriate laboratory instruments.	C	3
2	Analyze experimental data to verify fundamental physical laws and interpret the results.	C	4
3	Collaborate effectively in laboratory groups and communicate experimental findings through reports and discussions	A	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1					x					
CLO.2			x							
CLO.3						x				

GRADUATE ATTRIBUTES (GAs)

GA	Description
GA-3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
GA-5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
GA-6	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.

CSC 1208 Object Oriented Programming Techniques

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Object Oriented Programming Techniques	Semester: 2 Credits: 3
Course Code: CSC 1208	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 Hours/Week.
Corresponding Lab	Lab Course Name: Lab Object Oriented Programming Techniques
Course Code(If any): CSCL 1208	
Course Pre-Requisite: CSC 1103, CSCL 1103	

Course Objectives

The objectives of this course are to: 1. equip students with the required object-oriented programming skills required to build highly reusable, robust and maintainable software systems.

2. Evaluate the acquired skills via means such as group coding, individual projects, and impromptu programming challenge in C++. Upon successful completion of this course, the student will be able to:

(Knowledge based) 1. Explain the many concepts in Object- oriented programming paradigm; 2. Understand the history behind the Java technology, its features and strengths; 3. Understand and explain how Java achieves platform portability; 4. Understand the theory behind many programming structures, constructs, library codes exposed by the Java language; (Skills) 1. Prepare and configure computer systems for Java development; 2. Produce well-documented and elegant programs written in Java; 3. Use the object-oriented technique to analyze software problems

Course Description

This course emphasis the concepts of object oriented techniques used in developing computer based system. The topics are, Evolution of Object Oriented Programming (OOP); Object Oriented Concepts and Principles; Problem Solving in Object Oriented Paradigm; OOP Design: Decomposition into Objects, Class-Hierarchy Design for Modeling; Association; Aggregation; Composition; Definition of Classes; Subclasses; Inheritance; Method Overriding; Dynamic Dispatch: Definition of Method-Call; Sub-typing: Subtype Polymorphism, Implicit Up-casts, Notion of Behavioral Replacement (Subtypes acting like Super-types), Relationship between Sub-typing and Inheritance; Generic Types; Static and Dynamic Typing; Object-Oriented Idioms for Encapsulation: Privacy and Visibility of Class Members, Interfaces, Abstract Base Classes; Using Library Components such as Collection Classes and Iterators; Exception Handling; Events and Event Handlers; Canonical Uses such as GUIs, Mobile Devices; Using Reactive Framework; Externally-Generated Events and Program-Generated Events

CLO	Description	Domain	BT Level*
1	Describe the fundamental Concepts and principle of object-oriented programming.	C	2
2	Identify objects, classes and their relationships to design and construct moderately complex object-oriented solution.	C	2
3	Apply and analyze the concept of encapsulation, method overloading, inheritance and polymorphism.	C	3,4
4	Create useful software applications using strong OOP concepts with effective algorithmic strategies	C	6

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1				X						
CLO.2			X							
CLO.3				X						
CLO.4					X					

GA	Description
3.	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

TEXT/REFERENCE BOOKS

- Object Oriented Programming in C++, Robert Lafore (4E), 2001
- C++ How to Program, Deitel & Deitel (6E), 2007

SYLLABUS

Week	Topics
1	Review of Basic Programming Concepts, Control Structures, Functions, Arrays, Strings. Computer Languages, Introduction to Object Technology.
2	Structure Definitions, accessing Structure Members, structure within Structures, Structures and classes, OOP, OO Design and styles, Characteristics of OO Languages
3	Class Scope and accessing Class Members, Class Objects, Constructors and Destructors
4	Access Specifiers (Public, Private, Protected) in Class. Copy Constructor and dynamic memory management, Static Class Members
5	Encapsulation and Information Hiding, Objects as function arguments, Returning Objects from functions
6	Composition (Objects as members of Classes), Fundamentals of Operator Overloading, Restrictions On Operator Overloading ,Overloading Unary Operators
7	Overloading Binary Operators, Overloading stream operators, Introduction to Inheritance, Base Class, Derived Class, Protected Members, Scope Resolutions, Overriding Member functions ,Class Hierarchies
8	Mid Term
9	Public, Protected and Private Inheritance, Levels of Inheritance, Multiple Inheritance, Using 'this' pointer, 'const' objects and 'const' member functions.
10	Friend Functions, Friend Classes, Composition vs Inheritance, Data conversion
11	Introduction to Files, Steps of File Operations, Creating Random Access File, Writing and Reading from Random Access File
12	Introduction to Polymorphism, Relation among Objects in an Inheritance hierarchy, Polymorphism, Binding, Static and Dynamic Binding
13	Abstract Classes ,Virtual Functions with examples, Virtual Destructors
14	Introduction to Templates, Function Templates, Overriding Function Templates, Class Templates, Examples
15	Exception Handling Mechanism, try, throw & catch statements, Multiple catch statements, Re-throwing an Exception

CSCL 1208 Lab Object Oriented Programming Techniques

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Object Oriented Programming Techniques	Semester: 2 Credits: 1
Course Code: CSCL 1208	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code (If any): CSC 1208	Theory Course Name: Object Oriented Programming Techniques
Course Pre-Requisite: CSC 1103, CSCL 1103	

Course Description

This course emphasis the concepts of object-oriented techniques used in developing computer-based system. The topics are, Evolution of Object Oriented Programming (OOP); Object Oriented Concepts and Principles; Problem Solving in Object Oriented Paradigm; OOP Design: Decomposition into Objects, Class-Hierarchy Design for Modeling; Association; Aggregation; Composition; Definition of Classes; Subclasses; Inheritance; Method Overriding; Polymorphism, Generic Types; Static and Dynamic Typing; Object- Oriented Idioms for Encapsulation: Privacy and Visibility of Class Members, Interfaces, Abstract Base Classes; Using Library Components such as Collection Classes and Iterators; Exception Handling; Events and Event Handlers; Canonical Uses such as GUIs, Mobile Devices; Using Reactive Framework; Externally-Generated Events and Program-Generated Events

Course Objectives

The objectives of this course are to: 1. equip students with the required object-oriented programming skills required to build highly reusable, robust and maintainable software systems. 2. Evaluate the acquired skills via means such as group coding, individual projects, and impromptu programming challenge in C++. Upon successful completion of this course, the student will be able to: (Knowledge based) 1. Explain the many concepts in Object- oriented programming paradigm; 2. Understand the history behind the OOP technology, its features and strengths; 3. Understand the theory behind many programming structures, constructs, library; (Skills) 1. Prepare and configure computer systems for C++ development; 2. Produce well-documented and elegant programs written in C++; 3. Use the object-oriented technique to analyze software problems

TEXT/REFERENCE BOOKS

1. Object Oriented Programming in C++, Robert Lafore, 4th Edition.
2. C++ How to program, Deitel & Deitel, 10th Edition, Pearson Education, 2017.

CLO	Description	Domain	BT Level*
1	The better understating of underlying theory of programming languages	C	2
2	Identify bugs and conclude the output of the program.	C	2
3	Implement concept of object-oriented programming to construct a given program.	C	3
4	Create useful software applications using strong OOP concepts with effective algorithmic strategies	C	6
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1			X							
CLO.2			X							
CLO.3				X						
CLO.4					X					

GA	Description
3.	Identify and solve complex computing problems by reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Review of Basic Programming Concepts
2	Structure Definitions, accessing Structure Members, Nested Structures, Structures and classes
3	Class Scope and accessing Class Members, Constructors and Destructors
4	Access Specifiers, Objects and methods, Copy Constructor, Static Class Members
5	Encapsulation and Information Hiding, Objects as function arguments, Returning Objects from functions, Using 'this' pointer, 'const' objects and 'const' member Functions.
6	Composition, Operator Overloading, Unary Operators Overloading
7	Overloading Binary Operators, Overloading stream operators, Inheritance, Function Overriding, Scope Resolutions
8	Mid Term
9	Public, Protected and Private Inheritance, Multiple and Multi-Level Inheritance
10	Friend Functions, Friend Classes, Composition vs. Inheritance
11	File Processing/Handling
12	Polymorphism, Relation among Objects in an Inheritance hierarchy, Polymorphism, Binding, Static and Dynamic Binding
13	Abstract Classes, Virtual Function
14	Introduction to Templates, Function Templates, Overriding Function Templates, Class Templates, Examples
15	Exception Handling and Revision

CSC 2101 Communication & Presentation Skills

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Communication & Presentation Skills	Semester: 2 Credits: 3
Course Code: CSC 2101	Course Type: Core
Course Area/Domain: General Education	Contact Hours: 3 hours/Week
Corresponding Lab Course Code(If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 1102	

COURSE OBJECTIVES

The course aims to facilitate students in achieving superior public speaking skills to supplement their text book knowledge. During the span of 15 weeks, students will be exposed to delivering speeches and presentations and several speaking exercises to help them set and achieve benchmarks in public speaking. By giving them considerable practical speaking experience, the course helps students speak with confidence and power.

COURSE DESCRIPTION

The course is aimed at improving English language communication and presentation skills of students. With a multidimensional approach, the course enables the students to practice the use of English in everyday situations, building upon all four skills: listening, speaking, reading and writing. It prepares them to participate in seminars and discussions and make effective presentations, with an awareness of the audience and effective use of verbal and non- verbal communication. The course addresses the basic English language issues faced by the learners, while also aiming to foster in them, critical skills to develop a concise and clear argument, respond to others' comments and negotiate their own point of view persuasively. The course uses an interactive, participatory methodology, to engage learners' interest and boost their confidence to use English in everyday communication in formal and informal contexts.

TEXT/REFERENCE BOOKS

The Art of Public Speaking by Stephen Lucas (10th Edition)

CLO	Description	Domain	BT Level*
1	Identify and explain the basics of communication, its types, process, and importance.	C	2
2	Develop the skill of active listening and responding to audience	C	3
3	Prepare and deliver presentations using the principles of effective communication.	C	3,5

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1							X			
CLO.2							X			
CLO.3								X		

GA	Description
7.	Communicate effectively with the computing community about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
8.	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice

SYLLABUS

Week	Topics
1	Introduction to communication and presentation skills, Orientation to the course, Importance of communication
2	Process of communication, Noise and of communication its types mode.
3	Active Listening, Listening for specific information, Speaking Practice
4	Types of communication, Barriers in communication, Speaking Effective practice
5	Seven Cs of Effective communication, general Understanding
6	Individual presentations with instructors feedback
7	Individual presentations with instructors feedback
8	Midterm
9	Active listening, Listening for specific information (Documentaries)
10	Argumentative/Persuasive speeches
11	Importance of reading, Three steps in the reading process, Reading Comprehension
12	Public speaking, Debating, Role plays, dealing with stage phobia
13	Individual Presentations with instructors feedback
14	Individual Presentations with instructors feedback
15	Individual Presentations

CSC 1207 Digital Logic Design

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Digital Logic Design	Semester: 2 Credits: 2
Course Code: CSC 1207	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week
Corresponding lab Course Code(If any): CSCL 2103	Lab Course Name: Lab Digital Logic Design
Course Pre-requisite: Nil	

COURSE OBJECTIVES

To impart knowledge about digital logic and to gain the ability to design various digital circuits. Through the basic understanding of Boolean algebra and number systems it introduces the student to the fundamentals of combinational logic design and then to sequential circuits (both synchronous and asynchronous). Memory systems are also covered. Finally, the student is introduced to Register Transfer Logic design and the structured implementation of controllers and micro programmed computers.

COURSE Description

This course covers the basics of digital logic circuits and design. Through the basic understanding of Boolean algebra and number systems it introduces the student to the fundamentals of combinational logic design and then to sequential circuits (both synchronous and asynchronous). Memory systems are also covered. Finally, the student is introduced to Register Transfer Logic design and the structured implementation of controllers and micro programmed computers.

Text book

Floyd T.L, Digital Fundamentals, 10/e, Pearson Education, 2011
Mano M.M, Digital Design 5th Edition, Pearson Education
Fundamentals of Digital Logic with Verilog design, Stephen Brown, 2nd Edition

CLO	Description	Domain	BT Level*
1	Explain number systems, binary arithmetic, Boolean algebra, and logic gates, and analyze truth tables and standard forms of Boolean functions.	C	2,4
2	Apply Boolean minimization techniques (Algebraic, K-map, Quine-McCluskey) to design and simplify logic circuits	C	3,5
3	Design combinational and sequential logic circuits, including adders, subtractors, multiplexers, decoders, flip-flops, registers, and counters.	C	6
4	Demonstrate knowledge of digital memory devices, programmable logic devices (PAL, PLA, FPGA), and VHDL basics, and analyze their applications in digital systems.	C	2,4

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affectivedomain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				X						
CLO.4					X					

GA	Description
1.	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Introduction to digital system and different number systems, Binary arithmetic (addition, subtraction etc.)
2	Decimal to Binary Conversion, Binary to Decimal Conversion, Decimal to Octal Conversion, Octal to Decimal Conversion, Decimal to Hexadecimal Conversion, Hexadecimal to Decimal Conversion, Binary to Octal Conversion, Octal to Binary Conversion,
3	Duality and Truth tables, Standard forms of Boolean functions, Methods to obtain lists of SOP and POS forms of Boolean functions.
4	Effects of minimization, Boolean Minimization techniques, Algebraic method, Karnaugh map (K map) method, Don't care conditions. Quinn Mc- Cluskey method
5	Introduction to logic Gates, Derivation of Boolean expressions from logic circuits, Derivation of logic levels from logic circuits, Designing of logic circuits from Boolean expressions.
6	Introduction to Universal logic gates, Applications of logic gates, Introduction to logic conventions, Integrated circuits (ICs).
7	Introduction to combinational logic circuits, Procedure of designing combinational logic circuits, designing of custom built combinational logic circuits, Binary adders, and Parallel binary adders. Binary Arithmetic circuits: Ripple carry versus look ahead carry adders (Half adders, full adders), Binary Subtractor, Binary comparators.
8	Mid Term
9	Decoders (DECs), Encoders (ENCs), Multiplexers (MUXs), De-Multiplexers (De-MUXs), Error detecting circuits
10	Introduction to sequential logic circuits, Comparison of sequential and combinational logic circuits
11	Synchronous sequential logic circuits, Asynchronous sequential logic circuits.
12	Digital timing signal characteristics, Memory devices (Flip-Flops), Types of flip-flops, Characteristic and Excitation tables, Conversions among flip-flops, Memory designing with flip-flops. Introduction to register memory, Basic functions of shift registers, Types of shift registers, Shift register counters, Applications of shift registers.
13	Introduction to Finite State Machines, Moore Machines, Mealy Machines, Implementation of Finite State Machines. Asynchronous and Synchronous Mode Counters, Randomly Counting Circuits.
14	Memory Elements, Read and Write, Addressing, ROMs, PROMs and EPROMs, RAMs
15	Sequential Programmable Logic Devices - PAL, PLA, FPGA (Introduction and basic concepts only) Introduction to VHDL, Implementation of AND, OR, half adder and full adder.

CSCL 1207 Lab Digital Logic Design

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Digital Logic Design	Semester: 2 Credits: 1
Course Code: CSC 1207	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-requisite: Nil	

Course Description

This course covers the basics of digital logic circuits and design. Through the basic understanding of Boolean algebra and number systems it introduces the student to the fundamentals of combinational logic design and then to sequential circuits (both synchronous and asynchronous). Memory systems are also covered. Finally, the student is introduced to Register Transfer Logic design and the structured implementation of controllers and micro- programmed computers.

Course Objectives

To impart knowledge about digital logic and to gain the ability to design various digital circuits. Through the basic understanding of Boolean algebra and number systems it introduces the student to the fundamentals of combinational logic design and then to sequential circuits (both synchronous and asynchronous). Memory systems are also covered. Finally, the student is introduced to Register Transfer Logic design and the structured implementation of controllers and micro-programmed computers.

TEXT/REFERENCE BOOKS

1. Digital Logic and Computer Design M. Morris Mano, 3rd Edition, 2006.
2. Fundamentals of Digital Logic with VHDL Design, Stephen Brown 3rd Edition, 2006.

CLO	Description	Domain	BT Level*
1	Explain the basics of Multisim and demonstrate logic gate operations	C	2
2	Design basic arithmetic circuits using gates.	C	3
3	Verify and implement combinational circuits.	P	3
4	Implement sequential circuits and demonstrate basic FPGA implementation.	P	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1					X					
CLO.2				X						
CLO.3				X						
CLO.4				X						

GA	Description
4.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Introduction to Multisim & Trainer
2	Basics of Verilog Programming
3	To verify the operation OR, NOT, AND Gates
4	Verify the working of NAND , NOR Gates
5	Design AND, OR, NOT operations using Universal Gates NAND, NOR.
6	Verify the working of XOR, XNOR Gates
7	Design Half Adder Circuit, Full Adder, Half Subtractor Circuit & Full Subtractor
8	Mid Term
9	Verify the working of Multiplexer & Demux.
10	Designing one and two bit comparator
11	Design and Implementing Decoder and Encoder Circuit
12	Verify the working JK flip Flop (FF),RS flip flop (FF) and JK flip flop(FF)
13	Designing the Asynchronous Counters
14	Designing BCD counter
15	Introduction to FPGA

CSC 1209 Humanities Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Humanities	Semester: 1 Credits: 2
Course Code: CSC 1209	Course Type: Core
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Objectives

Develop a broad understanding of human culture, ethics, and philosophy.
Explore historical and contemporary perspectives on societal development.
Analyze the relationship between technology, ethics, and society.
Strengthen critical thinking and communication skills in social discourse.
Foster a sense of professional and ethical responsibility in AI applications.

Course Description

This course provides an introduction to the study of human culture, ethics, philosophy, and social sciences. It explores historical, cultural, and ethical perspectives that influence modern societies and technological advancements. Special emphasis is placed on ethical considerations in AI and computing, ensuring students understand their professional and social responsibilities.

Text/Reference Books:

Henry M. Sayre – Discovering the Humanities (Pearson, 2019)
Michael Sandel – Justice: What's the Right Thing to Do? (Farrar, Straus and Giroux, 2010)
Russ Shafer-Landau – The Fundamentals of Ethics (Oxford University Press, 2021)

Reference Books:

Yuval Noah Harari – Sapiens: A Brief History of Humankind (Harper, 2015)
Charles Bressler – Literary Criticism: An Introduction to Theory and Practice (Pearson, 2017)
John Rawls – A Theory of Justice (Harvard University Press, 2005)

SYLLABUS:

Week	Topics
1	Introduction to Humanities: Definition, Scope, and Importance
2	History of Civilization: Ancient to Modern Societies
3	Major Philosophical Thoughts: Greek, Islamic, and Western Philosophy
4	Ethics and Morality: Ethical Theories and Their Applications
5	Social Justice and Human Rights
6	Role of Culture and Language in Society
7	Media, Globalization, and Cultural Change
8	Midterm Exam
9	Science, Technology, and Society: Impact of AI and Automation
10	Professional Ethics in Computing and AI
11	Environmental Ethics and Sustainability
12	Leadership and Civic Engagement
13	Psychology and Human Behavior
14	Modern Social Issues: Gender Equality, Digital Privacy, and Cyber Ethics
15	Future of Humanities in the Age of AI and Digital Transformation

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Understand the fundamental concepts of humanities, ethics, and social sciences.	C	2
2	Analyze the impact of technology on society, culture, and ethics.	C	3
3	Apply ethical reasoning to real-world scenarios in AI and computing.	A	4

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	GRADUATE ATTRIBUTES (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2								X		
CLO.3									X	

GRADUATE ATTRIBUTES (GAs)

Gas	DESCRIPTION
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
8	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
9	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

CSC 1206 Probability and Statistics

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Probability and Statistics	Semester: 2 Credits: 3
Course Code: CSC 1206	Course Type: Core
Course Area/Domain: Mathematics & Science	Contact Hours: 3 hours/Week.
Foundation Courses	
Corresponding Lab Course Code(If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

A course “Statistics and Probability Theory” emphasizes the probabilistic foundations required to understand probability models and statistical methods. Topics covered will include the introduction to statistical methods (understanding data and its importance, data classification, tabulation, graphical representation), Central Tendencies, dispersion. Probability axioms, basic combinatory, discrete and continuous random variables, probability distributions, mathematical expectation, common families of probability distributions. Introduction to correlation and regression.

Course Objectives

After completing this course student should be able to demonstrate understanding of basic probability axioms and rules and the moments of discrete and continuous random variables as well as be familiar with common named discrete and continuous random variables and how to derive the probability density function of transformations of random variables and use these techniques to generate data from various distributions. Students will also be able to understand how to calculate probabilities, and derive the marginal and conditional distributions of bivariate random variables.

TEXT/REFERENCE BOOKS

Probability & Statistics for Engineers & Scientists, Ronald E. Walpole; Raymond H. Myers; Sharon L. Myers; Keying E. Ye 9th Edition Pearson, (2016) ISBN-13: 978-0134115856
 Elementary Statistics: A Step by Step Approach, Allan G. Bluman. 8th Edition, (2012)
 ISBN- 978-0-07-338610-2

CLO	Description	Domain	BT Level*
1	Introduction to the Statistical thinking, manipulating the data and organization specific emphasis to the computer sciences field like data mining.	C	2
2	Measure of Central Tendency, Location and measure of Dispersion. Defining the averages and possible dispersion in the data. Impact of the location of data on statistical analysis.	C	2,3,4
3	Understand uncertainty, model uncertain situations and find precise estimates using probability and its Laws	C	2,3
4	Understand Random Variables, discrete and continuous probability distributions, and their properties.	C	2,3
5	Applications of Continuous probability distribution in real life situations.	C	2,3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2		X								
CLO.3		X								
CLO.4		X								
CLO.5				X						

GA	Description
2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

SYLLABUS

Week	Topics
1	Introduction to Statistics and Data Analysis, Data Collection Methods, Populations, Samples, Sampling Procedures, Types of Statistical Studies
2	Measurement Scales, Random and Non-random Sampling, Tabulation, Presentation of data, Frequency Distribution,
3	Measures of central tendency (Arithmetic mean, Harmonic mean, and Geometric mean), Mode, Median, Quartiles, Deciles, Percentiles of ungrouped Data
4	Using Grouped Data to estimate Mean, Mode, Median, Quartiles, Deciles, and Percentiles.
5	Graphical representation of Data using Different Graphs and GA ts
6	Measure of Dispersion and their types, Skewness, coefficient of Variation-Consistency and relative comparison
7	Introduction to Simple Regression Model, Correlation coefficient and its interpretation to interpret relationships and modeling data.
8	Mid Terms
9	Introduction to set theory, Probability, Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules. Conditional Probability, Independence, and the Product Rule, Bayes' Rule. Application of probability in modeling of uncertainty.
10	Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, (Discrete and continuous).
11	Probability distributions (classification based on types of variable).
12	Bernoulli, Binomial, Poisson, Negative Binomial and geometric probability distributions.
13	Properties and their interpretations, Moment Generating Functions.
14	Normal distribution, Exponential distribution, Gamma distribution and their properties.
15	Application to the real life problems of Normal distribution, Exponential distribution and Gamma distribution

CSC 1209 Islamic Studies / Humanities

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Islamic Studies	Semester: 2 Credits: 2
Course Code: CSC 1209	Course Type: Core
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code(If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

The course begins with explanation of Articles of Islamic Faith including some discussion on Quran and modern science. See rah of Prophet (PBUH) is discussed in length as well as his missions as given in the holy Quran. Selected verses of Holy Quran are discussed followed by basics of science of Hadith and 20 selected a 'hadith. Worships like Prayers, Zakat, Fasting and Hajj are covered next. Last lectures are devoted to Islamic civilization, jurisprudence, economic system and the human rights.

Course Objectives

To provide essential information about Islamic Faith.
To improve skill of the students to perform prayers and other worships.
To enhance understanding of the students regarding Islamic Civilization, jurisprudence, economic system and human rights.

TEXT/REFERENCE BOOKS

Islamic Creed based on Quran and Sunnah / Author: Muhammad Bin Jameel Ar-Raheeq Al-Makhtoom / Author: Saif-ur- Rahman al Mubarakpuri Introduction to Islam / Author: Muhammad Hameed Ullah The Bible, the Quran and the Science / Author: Dr. Maurice Bucaille

CLO	Description	Domain	BT Level*
1	Demonstrate knowledge of all articles of Islamic faith and the compatibility of the Holy Quran with modern science.	C	1, 2
2	Understand Seerah of the Holy Prophet (PBUH), Ahadith and their most common classifications and Ibadaat in Islam.	C	2
3	Develop good concept of Islamic civilization, jurisprudence, economic system and the human rights.	C, A	2
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2								X		
CLO.3									X	

GA	Description
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
8	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
9	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

SYLLABUS

Week	Topics
1	Allah (SWT) His signs- Comparison of information given in the Holy Quran about the creation of universe with that of modern science
2	Allah (SWT), His Oneness and His relationship with the human beings as explained in the Holy Quran.
3	Prophet (PBUH), His Place and Mission according to the holy Quran and his own blessed words.
4	Prophet (PBUH), His Seerah, Meccan period, Madinah period, Some lessons to learn from Messenger Madina and Farewell Pilgrimage.
5	The Holy Quran, General information, Preservation, Recitation and our Duties. Selected Verses from 2nd and 4th Chapter.
6	The Holy Quran, Selected Verses from 6th, 9th, 25th, 39th, 49th 61st Chapters.
7	Hadith, Basics of the Science of Hadith. Most used classifications.
8	Mid Term, Selected Ahadith.(20).
9	Islamic Faith- Belief in Angels, Books, Prophets and the Judgment Day.
10	Worships in Islam- Salah and Zakat
11	Worships in Islam- Fasting and Hajj
12	Islamic Civilization- Historical developments, its characteristics and influence over humanity
13	Basics of Islamic Jurisprudence
14	The Islamic Economic System
15	Human rights in Islam

CSC 1201 Discrete Mathematical Structures

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Discrete Mathematical Structures	Semester: 3 Credits: 3
Course Code: CSC 1201	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Objectives

Introduces the foundations of discrete mathematics as they apply to Computer Science, focusing on providing a solid theoretical foundation for further work. Further, this course aims to develop understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through study of combinatorial reasoning, abstract algebra, iterative procedures, predicate calculus, tree and graph structures

Course Description

This course introduces fundamental mathematical structures necessary for the development of program logic. It covers logic: propositional logic; negation disjunction and conjunction; implication and equivalence; truth tables; predicates; quantifiers; Set theory: inductive definition of sets and proof by induction; Relations: representation of relations by graphs; properties of relations, equivalence relations Functions: mappings, injection and surjection, composition of functions; inverse functions; Recursive function theory; Elementary combinatoric; counting techniques; recurrence relation; generating functions. Graph Theory: elements of graph theory, Planar Graphs, Graph Coloring, Euler graph, Hamiltonian path, trees, spanning tree, minimal spanning tree and their applications.

TEXT/REFERENCE BOOKS

Discrete Mathematics and Its Applications, Kenneth H.Rosen, 6th Edition, McGraw Hill.

CLO	Description	Domain	BT Level*
1	Understand the key concepts of discrete structures such as proposition, sets, permutations, relations, graphs and tree	C	2
2	Apply formal logic proofs and/or informal logical reasoning to real world problems by comprehending and constructing mathematical arguments	C	2
3	Perform combinatory analysis to solve counting problems	C	3
4	Apply the knowledge and skills obtained to investigate discrete mathematical problems and solve other problems in computing	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2		X								
CLO.3		X								
CLO.4		X								

GA	Description
2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements

SYLLABUS

Week	Topics
1	Introduction to discrete structures, definitions, application in real world, important areas to cover. Logic definition, proposition (logic and variables), disjunction, conjunction, negations, conditional statements, bi-conditional statements
2	Truth tables for compound proposition, bit operations, translating English sentence into compound propositions, logical equivalences, Tautologies and contradiction, useful logical equivalences, Laws of logic, Notions of implication (converse, inverse, contrapositive)
3	Predicate and Quantifiers, Valid and Invalid argument form
4	Sets, representations of set, empty set, equal set, Venn diagram, subset, Power set, cardinality, ordered n-tuples, Cartesian product, set operations (union, intersection, complement), Computer representation of set
5	Relations, relations on a set, properties of relation, reflexive, symmetric relations, antisymmetric, transitive relations
6	Combining relations, Composition of relations, Matrix representation of relations, representing relations using digraphs, equivalence relation
7	Functions, domain, codomain, range, one-to-one functions, Onto function
8	Mid Term
9	Sequence, Series, Recursion, Composition of functions, one to one correspondence, equality of functions, identity function, inverse function
10	Mathematical induction, structural induction, strong and weak induction, Mathematical induction for divisibility
11	Combination, Permutations, Combinations, K-Combinations
12	Tree diagram, Inclusion-exclusion principle, Pigeonhole Principle
13	Graphs, Paths and circuits, Matrix representation, Isomorphism
14	Planar graphs, Trees, Spanning Trees
15	Boolean Algebra

CSC 2102 Data Structures and Algorithms

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Data Structure and Algorithms	Semester: 3 Credit: 3
Course Code: CSC 2102	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/week.
Corresponding Lab Course Code(If any): CSCL 2102	Lab Course Name: Lab: Data Structure and Algorithms
Course Pre-Requisite: CSC 1208, CSCL 1208	

Course Objectives

The purpose of this course is to provide the students with solid foundations in the basic concepts of programming data structures and algorithms. The main objective of the course is to teach the students how to select and design data structures and algorithms that are appropriate for problems that they might encounter. This course is also about showing the correctness of algorithms and studying their computational complexities.

Course Description

Introduction to Data Structures and Algorithms, Arrays, Linked Lists, Stacks, Queues, analysis of algorithms, trees, binary search trees, hash tables, graphs, graph traversals, graph algorithms, Sorting. Introduction to Data Structures and Algorithms, Arrays, Linked Lists, Stacks, Queues, analysis of algorithms, trees, binary search trees, hash tables, graphs, graph traversals, graph algorithms, Sorting.

TEXT/REFERENCE BOOKS

Data Structures and Algorithms in Java, fifth edition Michael T. Goodrich and Roberto Tamassia John Wiley & Sons Inc., 2010.
Data Structures and Algorithms in Java, 2nd Edition, Robert Lafore, Sams Publishing, 2003.

CLO	Description	Domain	BT Level*
1	Implement various data Structures and their algorithms and apply them in implementing simple applications.	C	2,3
2	Analyze simple algorithms and determine their complexities	C	4,5
3	Apply the knowledge of data structures to other applications domain.	C	3
4	Implement binary search tree and perform tree based operations.	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1					X					
CLO.2			x							
CLO.3				x						
CLO.4		X								

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Data structures, importance and types. Abstract Data Types. Time and Space Complexity (Big O).
2	Static vs Dynamic data structures, Arrays, Structures. Need for list. Implementing sequential lists using arrays. List Operations.
3	Singly, Doubly & Circular linked Lists and various operations on them.
4	Stacks and their importance. Array Representation. Stack Operations. Infix, Postfix and Prefix Expressions, Infix to postfix conversion Algorithm, Evaluating postfix expression.
5	Recursion: Function Call and Recursion Implementation. Recursion, Non-tail Recursion, Indirect Recursion, Nested Recursion, Backtracking.
6	Queues. Operations. Circular, Double Ended and Priority Queues. Applications of Queues.
7	Basic concepts of rooted Tree, Binary tree and tree traversal algorithms.
8	Binary Search trees and its various operations, linear search
9	AVL Trees. Insertion, Deletion and Rotations.
10	Building Max and Min Heaps. Deletion in a heap. Heap Sort.
11	Graph and the representations. Adjacency Matrix. Adjacency List. Directed and Undirected Graphs. Application area of graph. Traversal techniques. Minimum spanning trees.
12	Shortest path algorithm: Dijkstra's Algorithm, Bellman ford algorithm
13	Hash functions and tables. Strategies for avoiding and resolving collisions.
14	Binary search and its time complexity.
15	Bubble, Selection, Insertion, Merge and Quick Sorts.

CSCL 2102 Lab Data Structures and Algorithms

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Data Structures and Algorithms	Semester: 3 Credits: 1
Course Code: CSCL 2102	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code: CSC 2102	Theory Course Name: Data Structures and Algorithms
Course Pre-Requisite: CSC 1103, CSCL 1103	

Course Description

The main topics covered in this course are as follows Introduction, Arrays, Linked Lists, Stacks, Queues, analysis of algorithms, trees, binary search trees, hash tables, graphs, graph traversals, graph algorithms, Sorting.

Course Objectives

The purpose of this course is to provide the students with solid foundations in the basic concepts of programming data structures and algorithms. The main objective of the course is to teach the students how to select, design, and implement data structures and algorithms that are appropriate for problems that they might encounter. This course is also about showing the correctness of algorithms and studying their computational complexities. This lab course is designed is designed to implement of core concepts of data structures and algorithms.

TEXT/REFERENCE BOOKS

Data structures& Algorithms in C++, Michael T. Goodrich , Roberto Tamassisi, and David Mount, 2nd Edition, John Wiley ,2011

CLO	Description	Domain	BT Level *
1	Implement various data Structures and their algorithms and apply them in implementing simple applications.	C	2,3
2	Implement hash tables, including collision avoidance and resolution.	C	2

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1					X					
CLO.2			X							
CLO.3		X								

GA	Description
2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

SYLLABUS

Week	Topics
1	Java Programming Environment, Primitive Data Types
2	Single and 2D Implementation Array
3	Single Link List, Double Link List , Link List Operations
4	Stack Implementation , Push , POP, Stack Implementation with Link List
5	Queue Operations
6	Linear Search, Binary Search, Recursion
7	Binary Trees, Insertion / Display in Binary Trees, Tree Traversal (Preorder, Post order, In order) AVL Trees : Insertion , Deletion , Search and Traversal
8	Mid Term
9	Heap Implementation and Operations ; Min Heap, Max Heap
10	Graph , Breadth First Search , Depth First Search
11	Minimum Spanning Tree, Dijkstra's Algorithms
12	Hashing, Implementation of hash function, Hash tables, Insert, search and retrieve values from hash tables
13	Bubble Sort, Selection Sort and Its operations
14	Quick Sort, Merge Sort and Its operations
15	Insertion Sort Its operations and Complexity, Project

CSC 3105 Computer Organization and Assembly Language Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Computer Organization and Assembly Language	Semester: 3 Credits: 2
COURSECODE: CSC 3105	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code(If any): CSCL 2201	Lab Course Name: Lab Computer Organization and Assembly Language
Course Pre-Requisite: CSC 1207, CSCL 1207	

Course Objectives

The main objective of this course is to introduce the organization of computer systems and usage of assembly language for optimization and control. Emphasis should be given to expose the low-level logic employed for problem solving while using assembly language as a tool. At the end of the course the students should be capable of writing moderately complex assembly language subroutines and interfacing them to any high level language.

Course Description

This course covers the fundamental concepts of Computer organization and Assembly language. The topics include Octal and hexadecimal number systems, ASCII codes, Assembly language programming. Instruction formats and types, memory and I/O instructions, arithmetic instructions, addressing modes, stack operations, procedures, conditional processing, shift and rotate instructions along with Multiplication and division. Advance procedures and interrupt handling. Practice of assembly language programming.

TEXT/REFERENCE BOOKS

Assembly Language for x86 Processors, 6th Edition, Kip R. Irvine, Prentice Hall, 2010 Computer Organization and Design, 3rd Edition, David Peterson, Morgan Kaufman, 2005

CLO	Description	Domain	BT Level*
1	Discuss and write programming code for the basic arithmetic and logic operations available in assembly language	C	2,3
2	Write assembly language code demonstrating how 8, 16-bit values can be used.	C	2
3	Discuss and illustrate with programming code the use of branching, flags, stacks, procedures, and interrupts	C	2
4	Develop assembly programs to solve computational problems efficiently.	C	4

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2				X						
CLO.3			X							
CLO.4				X						

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

SYLLABUS

Week	Topics
1	Introduction to computer organization. Welcome to Assembly language; Assembly Language Applications; Virtual Machine Concept
2	Machine Level Representation of Data: Data representations, Bits, bytes, and words. Numeric data representation and number bases.
3	Fixed- and floating-point systems. Signed and twos-complement representations
4	Assembly Level Machine Organization: Basic organization of the von Neumann machine. Control unit; Instruction fetch, decode, and execution.,
5	Instruction sets and types (data manipulation, control, I/O), Basic elements of Assembly language. Assembling, Linking, and Debugging.
6	Assembly/machine language programming. x86 Instruction formats. x86 Addressing modes, Data Definitions. Symbolic Constants. Data Transfer Instructions.
7	Data-related Operators and Directives; ADD, SUB, INC, DEC, NEG, LOOP. Examples, Stack Operations, Subroutine call and return mechanisms, Boolean and Comparison Instructions; AND, OR Instructions with practice examples.
8	Mid Term
9	XOR & NOT Instruction; Applications; and practice programs for comparisons. TEST Instruction; CMP Instruction; comparison of these instructions, Nested Loops
10	Conditional Jumps: Types of Conditional Jumps; Jumps Based on Specific Flags, Equality, Unsigned Comparisons & Signed Comparisons; Conditional Jump Applications
11	Shift and Rotate Instructions; Logical vs. Arithmetic Shifts; SHL, SHR, SAL and SAR Instructions; ROL, ROR, Comparisons for shifts and rotate using programs,
12	RCL and RCR Instructions; SHLD/SHRD Instructions; Shift and Rotate Applications, Multiplication and Division Instructions: MUL, IMUL along with practice examples,
13	DIV Instruction; Signed extension instruction; IDIV Instruction; Implementing Arithmetic Expressions, Practice examples of MUL, IMUL, DIV and IDIV
14	Advanced Procedures: Local Variables, Stack Parameters, Passing parameters to a procedure, RET Instruction,
15	Interrupts & IO (Software and Hardware): Interrupts introduction: Hardware/Software interrupts, Interrupt handling routine, PC interrupt handling

CSCL 3105 Lab Computer Organization and Assembly Language

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab: Computer Organization and Assembly Language	Semester: 3 Credits: 1
Course Code: CSCL 3105	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code(If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 1207, CSCL 1207	

Course Objectives

The main objective of this course is to introduce the organization of computer systems and usage of assembly language for optimization and control. Emphasis should be given to expose the low-level logic employed for problem solving while using assembly language as a tool. At the end of the course the students should be capable of writing moderately complex assembly language subroutines and interfacing them to any high level language.

Course Description

This course covers the fundamental concepts of Computer organization and Assembly language. The topics include Octal and hexadecimal number systems, ASCII codes, Assembly language programming. Instruction formats and types, memory and I/O instructions, arithmetic instructions, addressing modes, stack operations, procedures, conditional processing, shift and rotate instructions along with Multiplication and division. Advance procedures and interrupt handling. Practice of assembly language programming.

TEXT/REFERENCE BOOKS

Assembly Language for x86 Processors, 6th Edition, Kip R. Irvine, Prentice Hall, 2010

CLO	Description	Domain	BT Level*
1	Acquire the basic knowledge of computer organization, computer architecture and assembly language.	C	2
2	Implement assembly programs involving data movement, arithmetic operations, and subroutines.	C	3
3	Debug and test assembly programs to verify correct hardware-level behavior.	C	4
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2				X						
CLO.3										
CLO.4						X				

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
6.	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.

SYLLABUS

Week	Topics
1	Introduction to Assembly Language, Microprocessors, Architecture (Intel 8086), How program execute?
2	MASM, NASM, Emulator (in detail)
3	Data Registers: AX, BX, CX, DX, Instructions: MOV, ADD, SUB, INC, DEC, Program structure, Basic I/O Instructions
4	Number Systems, Segment Offset Addressing Scheme, Segment Registers: CS, DS, SS, Memory Segment
5	Binary I/O, Hexadecimal I/O, Conversions, Count number of zeros & ones in a character
6	Variables, Data Declaration Directives: DB, DW, DD, DQ, DT, Memory to Memory Operation Issue, Control Registers: OF, DF, SF, ZF, AF, PF, CF
7	Flow Control, JMP Instruction, Unconditional Jumps, Conditional Jumps
8	Midterm Examination
9	Single Flag Jumps, LOOP Instruction
10	Logical Instructions, NOT, AND, OR, XOR, CMP Instruction, TEST Instruction
11	Shift Instructions, Rotate Instructions, MUL Instruction, DIV Instruction, Decimal Input and Output Procedures
12	Arrays, Addressing Modes, Index Registers: SI, DI
13	Stack, Pointer Registers: IP, SP, BP, Procedures
14	Interrupts (10h 00-0A, 21h)
15	Revision and practice / project

CSC3106 HCI and Computer Graphics

COURSE INFORMATION SHEET

PROGRAMME: BSCS/BSSE/BSAI	DEGREE: BSCS
COURSE: HCI and Computer Graphics	SEMESTER: III CREDITS: 2
COURSE CODE: CSC1111	COURSE TYPE: Domain Core
COURSE AREA/DOMAIN: Core	CONTACT HOURS: 2 hours/Week
CORRESPONDING LAB COURSE CODE(IFANY): CSCL 3106	LAB COURSE NAME: Lab: HCI and Computer Graphics

COURSE OBJECTIVES:

This course will enable the student to understand fundamental HCI principles and apply usability heuristics in interface design, analyze cognitive and perceptual models that influence user interaction, apply 2D and 3D graphics techniques, including color theory, visual hierarchy, and rendering and evaluate and improve user interfaces through prototyping and usability testing.

COURSE DESCRIPTION:

This course provides an introduction to Human-Computer Interaction (HCI) and Computer Graphics, focusing on designing and evaluating user-friendly interfaces and understanding graphical representations in computing. The HCI component explores user-centered design principles, usability heuristics, cognitive models, and accessibility considerations. The Computer Graphics component introduces 2D and 3D graphics, rendering techniques, color theory, and visual hierarchy, emphasizing their role in interface design and interaction.

SYLLABUS:

Week	Topics
1	Contexts for HCI. Norman's Design Principles: Affordances, Natural Mapping, Constraints, Feedback, Visibility, Conceptual Model + Introduction to Computer Graphics
2	User-Centered Design: Personas and Scenarios + Fundamentals of 2D Graphics (Raster vs Vector, Coordinate Systems)

3	Usability Heuristics and Principles of Usability Testing + Color Theory & Typography in UI/Graphics
4	Cognitive Models that Inform Interaction (Memory & Human Error) + Motion and Attention in UI/Graphics (Animations, Transitions)
5	Gestalt Laws of Perception & Social Models in Interaction Design + Principles of Visual Hierarchy in Graphics
6	Principles of Good UI Design & Tradeoffs + Graphics Rendering Pipeline Overview
7	Accessibility (Design for Disabilities, Age Groups) + Color Contrast & Readability in Computer Graphics
8	Principles of GUI & Visual Design + User Needs Analysis
9	User Analysis & Task Analysis (HTA, GOMS) + Advanced Graphics Transformations
10	Prototyping (Low, Medium, High Fidelity) + UI Standards, Internationalization
11	Evaluation Without Users (Heuristic, Cognitive Walkthrough) + Basics of 3D Graphics (Modeling & Projections)
12	Evaluation With Users (Usability Testing, Ethics) + Data Visualization Techniques (Tree Maps, Table Lens)
13	Usability Inspection Methods, Human/System Failure & Social Engineering + Error Handling in UI & Graphics
14	Information Visualization Techniques & New Interaction Technologies (Touch, Multi-Touch) + 3D Graphics & AR/VR Basics
15	Speech Recognition, NLP, Ubiquitous Computing + Augmented & Virtual Reality

TEXT/REFERENCE BOOKS:

1. **Norman, D. A.** (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books.
2. **Shneiderman, B., & Plaisant, C.** (2017). *Designing the User Interface: Strategies for Effective Human-Computer Interaction* (6th ed.). Pearson.
3. **Dix, A., Finlay, J., Abowd, G., & Beale, R.** (2004). *Human-Computer Interaction* (3rd ed.). Pearson.
4. **Hughes, J. F., van Dam, A., McGuire, M., Sklar, D. F., Foley, J. D., Feiner, S. K., & Akeley, K.** (2013). *Computer Graphics: Principles and Practice* (3rd ed.). Addison-Wesley.

COURSE LEARNING OUTCOMES (CLOs):

CLO	DESCRIPTION	Domain	BT Level
1	Understand fundamental HCI principles and their role in designing user-friendly interfaces.	Cognitive	2, 3
2	Apply usability heuristics and cognitive models to improve interaction design.	Cognitive	2, 3
3	Use basic 2D and 3D graphics concepts, including color,	Cognitive	2

	typography, and rendering, in UI/UX.		
4	Develop and evaluate interactive prototypes through usability testing methods.	Cognitive	2

Graduate Attributes:

GA	Description
2	Apply computing fundamentals, design principles, human–computer interaction concepts, and cognitive psychology to understand and conceptualize user-centered systems.
3	Identify usability, accessibility, and interaction problems and propose substantiated solutions using HCI principles, visual design theory, and cognitive models.
4.	Design, prototype, and evaluate interactive systems, interfaces, and visual graphics elements that meet user needs and usability standards.
5	Select and use modern prototyping tools, graphics frameworks, and usability evaluation methods with awareness of their capabilities and limitations.

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3					X					
CLO.4				X						

CSCL3106 Lab:HCI and Computer Graphics

COURSE INFORMATION SHEET

PROGRAMME: BSCS/BSSE/BSAI	DEGREE: BSCS
COURSE: Lab: HCI and Computer Graphics	SEMESTER: III CREDITS: 1
COURSE CODE: CSCL1111	COURSE TYPE: Domain Core
COURSE AREA/DOMAIN: Core	CONTACT HOURS: 3 hours/Week
CORRESPONDING COURSE CODE(IFANY): CSC 3106	COURSE NAME: HCI and Computer Graphics

COURSE OBJECTIVES:

By the end of this course, students will be able to:

1. Understand HCI principles and their role in user interface design.
2. Apply usability heuristics and user-centered design techniques.
3. Implement basic 2D and 3D graphics concepts for UI and visualization.
4. Evaluate and improve UI design through prototyping and usability testing.

COURSE DESCRIPTION:

This course covers Human-Computer Interaction (HCI) and Computer Graphics, focusing on usability principles, user-centered design, visual hierarchy, and interface evaluation. It introduces 2D and 3D graphics, animations, rendering techniques, and UI prototyping to enhance interaction and user experience.

SYLLABUS:

Week	Topics
1	Setting up development environment for GUI and graphics programming
2	Designing wireframes for simple applications
3	Drawing basic shapes, colors, and transformations in 2D
4	Implementing UI elements and handling user interactions
5	Applying usability heuristics to a simple UI

6	Implementing 3D transformations in graphics programming
7	Basic rendering techniques and applying lighting effects
8	mid exams
9	Creating simple animations and transitions
10	Implementing multi-touch gestures in an application
11	Designing a simple web/mobile UI prototype
12	Conducting usability testing on student-designed UIs
13	Hands-on project development session
14	Final project implementation and debugging
15	Final project presentations

TEXT/REFERENCE BOOKS:

- **Norman, D. A.** (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books.
- **Shneiderman, B., & Plaisant, C.** (2017). *Designing the User Interface: Strategies for Effective Human-Computer Interaction* (6th ed.). Pearson.

COURSE LEARNING OUTCOMES (CLOs):

CLO	DESCRIPTION	Domain	BT Level
1	Implement user interface components by applying fundamental HCI principles and usability guidelines using a prototyping tool.	Cognitive	3
2	Apply usability heuristics and cognitive models to analyze and improve the usability of interface designs in practical lab tasks.	Cognitive	4
3	Create and manipulate 2D/3D graphics elements—such as colors, typography, and visual layouts—to develop visually effective user interface screens.	Cognitive	3
4	Develop and evaluate interactive prototypes through user testing techniques, documenting findings and recommending improvements.	Cognitive	5

CSC 1211 Ideology and Constitution of Pakistan Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Ideology and Constitution of Pakistan	Semester: 3 Credits: 2
Course Code: CSC 1211	Course Type: Core
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code(If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

This course introduces the history of Pakistan with reference to pre- and post- independence eras, and the contribution of different governments in social, economic and legislative development over years. The second part of the course focuses on Islam and the present-day Muslim world

Course Objectives

Develop vision of historical perspective, government, politics, contemporary Pakistan, ideological background of Pakistan. • Study the process of governance, national development, issues arising in the modern age and posing challenges to Pakistan.

TEXT/REFERENCE BOOKS

Pakistan Studies / Author: M. R. Kazimi / Year: 2011

CLO	Description	Domain	BT Level*
1	Create awareness among students about Pakistan as an enlightened nation by comparing it with the rationale and endeavors for Pakistan's creation	C, A	2
2	Educate students about the key concepts in disciplines comprising Pakistan Studies, including history, economics, and political science	C	2
3	Assist students in identifying various perspectives on current, persistent, and controversial issues in Pakistan; identify their own position on these issues, and be able to support them	C	2

	Graduate Attributes (GAs)										
	1	2	3	4	5	6	7	8	9	10	
CLO.1	X										
CLO.2			X								
CLO.3			X								

GA	Description
1	Students will analyze and compare historical and ideological foundations of Pakistan through written assignments, reflections, or quizzes that assess their understanding of the rationale behind Pakistan's creation and its vision as an enlightened nation.
3.	Students will participate in debates, position papers, or analytical discussions to demonstrate the ability to identify differing perspectives on Pakistan's contemporary issues and support their own viewpoint with evidence.

SYLLABUS

Week	Topics
1	Historical background of creation of Pakistan Two Nation Theory Downfall of Mughal Empire & War of Independence 1857
2	Sir Syed Ahmed Khan & Educational Movements Ideology of Pakistan, Pakistan Movement: A Historical Prospect (Birth of Muslim Nationalism)
3	Creation of Pakistan as an Independent State: Early Problems
4	Constitutional development & Political History (1) 1947-1958
5	Constitutional development & Political History (2) 1971-1977 1977- 1958-1971 1988
6	Constitutional development & Political History (3) 1988-1999
7	Constitutional development & Political History (4) 1999 onwards
8	Mid Term
9	Lessons From History
10	Current Issues in Pakistan Tolerance and empathy Human rights Other social issues
11	Foreign Policy
12	Pakistan and the World South Asia China US Middle East
13	Future Outlook Science and technology Education
14	Economy, Development and Industry
15	Healthcare Employment Rights

CSC 2203 Database Systems

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Database Systems	Semester: 4 Credits: 3
Course Code: CSC 2203	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code(If any): CSCL 2203	Lab Course Name: Lab Database Systems
Course Pre-Requisite: CSC 2102, CSCL 2102	

Course Description

This is an introductory course to database, its main aim is to introduce database to students as the heart of most business applications. The course focuses on database design, modeling and database language SQL.

Course Objectives

Introduce the concept of DBMS and its purpose. Introduce the main concepts in a Database (Tables, records, fields, primary key, foreign key, relation etc). Learn how to design a logical database model, convert the logical database designs to physical designs (Mapping), and develop the physical database. Introduce the concept of normalization theory (1NF, 2NF, and 3NF). Enable students to build a desktop application with a database. Enhance students skills including: team working, problem-solving, and self-learning.

Text book

Database Systems Models, Languages, Design, and Application Programming. 6th edition, Ramez Elmasri & Shamkant Navathe.

CLO	Description	Domain	BT Level*
1	Understand database concepts and applications, Understand data models, schemas and instances, storage organizations concepts, database languages and interfaces and the database system environment	C	2
2	Implement the relational database design and data modeling using entity relationship (ER) model.	C	5
3	Implement SQL: Data definition, constraints, schema, queries and operations in SQL.	C	2
4	Produce well-structured database using functional dependencies and normalization, Build, design and tune databases.	C	4
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		x								
CLO.2				X						
CLO.3			X							
CLO.4										x

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Introduction to Databases: What is a Database/ DBMS? Characteristics of the Database Approach. When to/not to use a Database? Actors on the Scene.
2	Overview of Database Languages and Architectures: - Data Models, Schema, and Instances. Three Schema Architecture. - Data Independence. - DBMS Component Module. - DBMS Architectures.
3	Conceptual Data Modeling Using Entities and Relationships: Entity Types, Entity Sets, Attributes, and Keys. Relationship Types, Relationship Sets, and Multiplicity.
4	Weak Entity Types. ER Diagrams, Naming Conventions, and Design Issues. Relationship Types of Degree Higher than two. Enhanced ERD, Case Samples.
5	Mapping a Conceptual Design into a Logical Design:, Relational Database Design Using ER-to-Relational Mapping
6	Relational Model: Relational Model Concepts, Relational Model Constraints and Relational Database Schemas, Update Operations, Transactions, and Dealing with Constraints Violations.
7	Relational Algebra: Introduction., Select, Project, and Rename, Relational Algebra Operations from Set Theory: Union, Intersection, and Minus, Cartesian Product Operation
8	Join and its types: Inner Join, Outer Join (Left, right and full), Aggregate Functions and Grouping, Examples of Queries in Relational Algebra
9	SQL: Create Table, Data Types, Basic SQL Queries, Renaming ,Substring Pattern Matching and Arithmetic Operators, Insert, Delete, and Update
10	IS Null, Like, Between, IN, Order By, Advanced SQL: Nested Queries, Join, Aggregate Functions, Group By and Having Clauses
11	Database Design Theory (Normalization): Informal Design Guidelines for Relation Schemas, Functional Dependencies: Fully, partial dependencies, transitive dependency, Normal Forms Based on Primary Keys
12	General Definitions of Second Normal Form, Third Normal Forms Advantages and disadvantages of Normalization
13	Overview of Storage and Indexing, Types of indexing: Primary, secondary, clustered indexing, disk and file organization
14	Overview of Transaction Management, Concurrency Control and Recovery, Backup, types of backups: Full database backup
15	Differential backup, Transaction log backup, File or File group backup, And Review

CSCL 2203 Lab Database Systems

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Database Systems	Semester: 4 Credit: 1
Course Code: CSCL 2203	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3hours/Week.
Corresponding Theory Course Code: CSC 2203	Theory Course Name: Database Systems
Course Pre-Requisite: CSC 2102, CSCL 2102	

Course Description

This is the lab course of Database Systems, its main aim is to introduce database to students as the heart of most business applications. The course focuses on database design, modeling and database language SQL.

Course Objectives

The main objective of this course is to understand user requirements/views analyze existing and future data processing needs, develop an enterprise data model that reflects the organization & fundamental business rules. As a lab course students are introduced with entities, relationships, attributes, and business rules, integration and database views, normalization techniques, data integrity and security requirements. After Conceptual design we implement database using SQL queries and apply different operations constraints on Database.

TEXT/REFERENCE BOOKS

Database Systems Models, Language, Design and Application Programming, Ramez Elmasri and Shamkant Navathe, 6th Edition.

CLO	Description	Domain	BT Level*
1	Understand database concepts and applications, Understand data models, schemas and instances, storage organizations concepts, database languages and interfaces and the database system environment	C	2
2	Implement the relational database design and data modeling using entity relationship (ER) model.	C	5
3	Implement SQL: Data definition, constraints, schema, queries and operations in SQL.	C	2
4	Produce well-structured database using functional dependencies and normalization, Build, design and tune databases.	C	4

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2				X						
CLO.3			X							
CLO.4										X

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Introduction to MySQL -Installation and configuration
2	MYSQL tool description and detail
3	Create ER model using MySQL Data Modeling
4	Various Example of ER-Diagram with case study
5	Map ER model into relational Model
6	Practical view of relation, database schema -Different operations, transition, constraints on data
7	Create table -Insert , update and delete data -Basic retrieval queries using Select, from and where clause -Delete table and columns -Alter table
8	Use of JOIN in SQL and its types
9	Creation of table with attribute constraints -Key and referential integrity (Foreign key) constraints -Constraints on tuple using CHECK
10	Retrieving specific columns -Select distinct rows, Top rows, Like, IN, Between And operators -Logical operators -Ordering record
11	Generating sub queries -Aggregate functions -Group by and having clause
12	Introduction to front end tool -Understanding GUI
13	Connecting to database -Adding recording using hard coded SQL
14	Add, delete, and update a new record using forms -Record navigation using Forms - Reports and filters
15	Project Demo

CSC 3101 Computer Architecture Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Linear Algebra	Semester: 4 Credits: 3
Course Code: CSC 2206	Course Type: Core
Course Area/Domain: Maths	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

This course introduces students to the fundamental concepts and design principles of modern computer systems. The course covers the organization and structure of computer hardware, instruction set architecture (ISA), CPU design, memory hierarchy, input/output systems, and emerging trends in multicore and parallel architectures. Students will gain a strong understanding of how hardware components interact to execute programs efficiently, as well as practical insights into modern processor architectures, including RISC-V. The course emphasizes both theoretical knowledge and hands-on analysis of computer architectures, preparing students for advanced topics in system design, optimization, and performance evaluation.

Course Objectives

By the end of this course, students will be able to:

1. **Understand foundational concepts** in computer architecture and organization, including the Von Neumann model, data representation, and number systems.
2. **Explain instruction set architecture (ISA) concepts**, instruction formats, addressing modes, and basic assembly programming.
3. **Analyze CPU components** such as ALU, registers, buses, and understand the role of the control unit (hardwired and microprogrammed).
4. **Describe pipelining techniques** and identify pipeline hazards, including solutions like forwarding and branch prediction.
5. **Explain memory hierarchy** concepts, cache mapping, replacement policies, and virtual memory organization.

TEXT/REFERENCE BOOK

Computer Architecture: A Quantitative Approach — 6th Edition, by John L. Hennessy & Computer Organization and Design: The Hardware/Software Interface — RISC-V Edition,

CLO	Description	Domain	BT Level*
1	Explain the key components and operational principles of computer systems including ISA, datapath, control, memory hierarchy, and I/O subsystems.	C	2
2	Analyze processor and memory performance using CPI, pipeline behavior, and cache efficiency metrics	C	4
3	Design and evaluate simple processor datapaths, pipelined structures, or memory subsystems based on given design requirements	C	5
4	Use modern architectural tools or simulators to model system behavior and effectively document or present results	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				X						
CLO.4							X			

GA	Description
2	Understanding of Key Computer Architecture Concepts: Evaluate the student's comprehension of computer system components, including instruction set architecture (ISA), datapath, control unit, memory hierarchy, and I/O subsystems, through written exams, quizzes, or conceptual assignments.
3	Analysis of Processor and Memory Performance: Assess the ability of students to analyze processor behavior and memory performance using metrics such as CPI (Cycles per Instruction), pipeline efficiency, cache hit/miss rates, and memory access times. Assessment can be through problem-solving exercises, performance calculations, or case studies.
4	Design and Evaluation Skills: Measure students' capability to design simple processor datapaths, pipelined structures, or memory subsystems based on given specifications. Students demonstrate skills through design exercises, simulations, or structured design reports.
7	Practical Application and Documentation: Evaluate the use of modern architectural tools, simulators, or modeling software to simulate system behavior, collect results, and present findings effectively. Assessment can include lab reports, project presentations, or simulation-based assignments.

SYLLABUS

1	Introduction to Computer Architecture & Organization
2	Von Neumann Model, Number Systems, Data Representation
3	Instruction Set Architecture (ISA) Concepts
4	Instruction Formats, Addressing Modes, Basic Assembly
5	CPU Datapath Components (ALU, Registers, Buses)
6	Control Unit — Hardwired & Microprogrammed
7	Introduction to Pipelining
8	Pipeline Hazards, Forwarding, Prediction Techniques
9	Memory Hierarchy Basics
10	Cache Mapping, Replacement Policies, Virtual Memory
11	I/O Systems, Interrupts
12	DMA, Bus Architecture, Peripheral Communication
13	Flynn's Taxonomy & Parallel Processing
14	Multicore, GPU Architecture, SIMD/MIMD
15	RISC-V Basics, Modern Architectures & Trends

CSC 2206 Linear Algebra Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Linear Algebra	Semester: 4 Credits: 3
Course Code: CSC 2206	Course Type: Core
Course Area/Domain: Maths	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

The objective of the course is to provide a rigorous approach towards the solutions of linear models, which involves more than one variable. The techniques discussed in this course can be implemented on a wide range of applications from physical world. The matrix algebra will be helpful in performing and understanding of matrix computations on a machine. The eigenvalues, eigenvectors, inner product spaces, orthogonality are useful concepts for the analysis of dynamical systems.

Course Objectives

This course has been designed to acquire a good understanding of the concepts and methods of linear algebra, to develop the ability to solve problems using the techniques of linear algebra using Eigen values and Eigen vectors.

TEXT/REFERENCE BOOKS

Elementary Linear Algebra by Howard Anton
Linear Algebra and its Applications by Gilbert Strang

CLO	Description	Domain	BT Level*
1	Apply matrices, determinants, and linear systems to computing problems	C	3
2	Analyze vector spaces, spanning sets, bases, and dependence	C	4
3	Evaluate inner products, eigenvalues, diagonalization, and transformations for CS models	C	5

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				X						

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

Syllabus

Week	Chapters	Session Topic
1	Applications of Linear Algebra in Computer Science, Matrices and Determinants	(i) Types of matrices (ii) Matrix multiplication (iii) Determinants
2	Systems of Linear Equations	(i) Types of linear systems of equations, homogeneous and non-homogeneous linear systems of equations. (ii) Possible solutions for homogeneous linear systems of equations. (iii) Possible solutions for non-homogeneous linear systems of equations. (iv) Solution of linear systems of equations by Gaussian elimination method and Gauss-Jordan elimination method.
3	Systems of Linear Equations	(i) Rank, invertibility and row echelon form of matrices. (ii) Inverse of matrix by Gauss-Jordan method.
4	Vector Spaces	(i) Euclidean vector spaces. (ii) Matrix space. (iii) Polynomial space. (iv) Continuous function space. (v) Sub-spaces of a vector space.
5	Vector Spaces	(i) Linear combination in a vector space. (ii) Spanning set of a vector space.
6	Linear Dependence	Conditions for linear dependence of vectors.
7	Linear Independence	Conditions for linear independence of vectors.
8	Midterm Exam	Midterm Exam
9	Bases	Method for finding the basis and dimension of a vector space.
10	Inner Product Space	(i) Euclidean inner product space. (ii) Inner product on matrix space. (iii) Inner product on polynomial space. (iv) Inner product on continuous function space.
11	Inner Product Space	(i) Conversion of arbitrary basis into orthogonal basis by Gram-Schmidt process. (ii) Conversion of orthogonal basis into orthonormal basis.
12	Eigen Values and Eigen Vectors	Eigen values, Eigen vectors and characteristic equations.
13	Eigen Values and Eigen Vectors	(i) Diagonalization (ii) Orthogonal Diagonalization
14	Quadratic Form	(i) Converting a quadratic form to a diagonal form. (ii) Reducing a quadratic form to a sum of squares. (iii) Rotation and translation of conics.
15	Linear Transformation	Linear transformation in finite dimensional vector spaces.

CSC 3202 Design and Analysis of Algorithms

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Design and Analysis of Algorithm	Semester: 4 Credits: 3
Course Code: CSC 3202	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 2102, CSCL 2102	

Course Objectives

The main objective of this course is to make the student familiar with subjects concerning algorithm complexity. By the end of the semester, the student should be able to:

- (1) Recognize the use of several design techniques (greedy, divide-and-conquer, dynamic programming) and use these methods to solve simple Problems.
- (2) write and solve recurrence relations for recursive algorithms.
- (3) Determine asymptotic growth rates for algorithms.
- (4) Prove correctness of simple algorithm

Course Description

This course applies design and analysis techniques to numeric and nonnumeric algorithms which act on data structures. Design is emphasized so that the student will be able to develop new algorithms. Analysis of algorithms is concerned with the resources an algorithm must use to reach a solution. Topics include introduction to algorithm, asymptotic complexity, sorting and searching, divide and conquer, greedy graph algorithms, dynamic programming, data compression, backtracking, branch and bound

TEXT/REFERENCE BOOKS

Introduction to Algorithms, Thomas H. Cormen Charles E. Leiserson Ronald L. Rivest Cliff, 3rd Edition, MIT Press, 2009

CLO	Description	Domain	BT Level*
1	Explain fundamental algorithmic concepts such as time complexity, space complexity and algorithmic design strategies.	C	2
2	Apply the algorithms and design techniques to solve problems.	C	3
3	Analyze the correctness and efficiency of algorithms for classical problems in various domains.	C	4

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1				X						
CLO.2		X								
CLO.3			X							

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

SYLLABUS

Week	Topics
1	What are Algorithms, Why do you need to study Algorithm, What kinds of problems are solved by Algorithms, Algorithms as Technology, correctness and generality of Algorithms, Major Factors in Designing an Algorithm
2	Pre-Condition, Post Condition, Partial Correctness of an Algorithm, Total Correctness of An Algorithm Loop Invariant, Correctness of an Iterative Algorithm
3	Correctness of Recursive Algorithm, Mathematical Induction, Growth Rate of functions, Asymptotic Notations
4	Major Assumptions in analyzing an algorithm, RAM Model Mathematical analysis of non-recursive algorithms, Complexity Classes
5	Time and Space Tradeoffs in algorithm, Brute force Algorithms and their Analysis
6	Mathematical Analysis of recursive algorithms, Substitution Method, Tree Methods
7	Master Theorem for solving recurrences, Divide and conquer Algorithm,
8	Mid-Exam+ Merge Sort and its Worst case analysis,
9	Best and Average Case Analysis of merge sort, Quick Sort and its Worst, Best and Average Case Analysis, Heap Sort
10	Dynamic Programming, elements of Dynamic Programming, Comparison of Dynamic Programming with divide and Conquer , Knapsack Problem solving using Dynamic Programming
11	Longest Common Subsequent problem solving through Dynamic Programming, Shortest Path finding through Dynamic Programming
12	Kruskal Algorithm, Dijkstra's Algorithm
13	Optimization/Approximation Algorithms
14	Graph and its types
15	Presentations

CSC 3107 Operating Systems Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Operating Systems	Semester: 5 Credits: 2
Course Code: CSC 3107	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week
Corresponding Lab Course Code (If any): CSCL 2205	Lab Course Name: Lab Operating Systems
Course Pre-Requisite: None	

COURSE OBJECTIVES

The purpose of this course is to provide an understanding of operating systems to students so they will be able to understand the terminology surrounding and describing operating systems. Students will become familiar with the kinds of abstractions provided by general purpose operating systems that facilitate the development of user applications. Also they will become familiar with the typical internal policies and mechanisms implemented by operating systems and how they impact system performance.

COURSE DESCRIPTION

This course is an introduction to the theory and practice behind modern operating systems. Topics will include what an operating system does, system calls and interfaces, processes, concurrent programming, resource scheduling and management, virtual memory, deadlocks and security.

TEXT/REFERENCE BOOKS

Operating systems concepts, Abraham Silberschatz, 9th Edition, Wiley, 2012.
Modern Operating Systems, Andrew S. Tanenbaum, 4th Edition, Pearson, 2014.

CLO	Description	Domain	BT Level*
1	Understand the characteristics of different structures of the operating systems and identify the core functions of the operating system.	C	2
2	Analyze and evaluate the algorithms of the core functions of the operating systems and explain the major performance issues with regard to the core functions	C	4, 5
3	Demonstrate the knowledge in applying system software and tools available in modern operating system.	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1				x						
CLO.2			x							
CLO.3					x					

GA	Description
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Operating system basics, system calls
2	Process Concept and scheduling, process scheduling algorithms
3	Multiprocessor scheduling
4	Inter-Process Communication
5	Multithreaded programming, multithreading models, threading issues, thread scheduling
6	Synchronization, critical section
7	Synchronization hardware, Synchronization problems, Deadlocks, detecting and recovering from deadlocks
8	Mid Term
9	Memory Management, swapping, contiguous memory allocation.
10	Segmentation & Paging, virtual memory management, demand paging, thrashing
11	Memory-mapped files, file systems, file concept
12	Directory and disk structure, directory implementation
13	Free space management, disk structure, directory implementation
14	Virtual Machines
15	System Protection, Operating system security

TEXT/REFERENCE BOOKS

Operating systems concepts, Abraham Silberschatz, 9th Edition, Wiley,2012. Modern Operating Systems, Andrew S. Tanenbaum, 4 th Edition, Pearson,2014.

CSCL 3107 Lab Operating Systems Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Operating Systems	Semester: 5 Credit: 1
Course Code: CSCL 3107	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code: CSC 2205	Theory Course Name: Operating Systems
Course Pre-Requisite: None	

Course Description

This is a lab course to operating system, its main aim is to introduce operating system functionality to students. The lab course based on LINUX operating system. In lab students are given practical overview of OS concepts.

Course Objectives

The purpose of this course is to conduct parallel lab sessions against the Theoretical Operating System classes. Whatever topics students are being taught in the Theory classes, some/most of those are supposed to be covered in these equivalent lab sessions in order to simulate those concepts programmatically and subsequently to provide a better understanding of the Theoretical course.

TEXT/REFERENCE BOOKS

Operating system concepts, Galvin, Peter B., Greg Gagne, and Abraham Silberschatz, null Edition, John Wiley & Sons, 2016

CLO	Description	Domain	BT Level*
1	The better understating the underlying theory of programming languages	C	3
2	Enable a student to choose the appropriate Language for a Project	C	6
3	Learning of formal semantics design for a programming language	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2		X								
CLO.3			X							

GA	Description
1.	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problem.
3.	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

SYLLABUS

Week	Topics
1	Operating systems overview, Introduction to Linux (Ubuntu Installation)
2	Introduction to Linux commands
3	How to write C program in Linux, Implementation the Process Management using fork
4	Implementation the Process Management using wait and exit
5	Inter process communication and implementation through pipes, Implementation of Threads
6	Execution of Multithreading Program, Implementation of Lock with and without busy-waiting
7	Implement the Producer & consumer Problem (Mutex-Semaphore)
8	Mid Term
9	Implement Deadlock creation and avoidance, Simulating Banker's Algorithm for Deadlock Avoidance, Implement Process scheduling algorithm; FIFO, SJF, RR and PPS, Simulating multi-level queue scheduling algorithm.
10	Implement Deadlock Detection Algorithm in C
11	Implementation of Memory Management; allocation, resizing, releasing, clean
12	Simulation of Paging technique of Memory Management
13	Simulate page replacement algorithms : FIFO, LRU, LFU
14	Implementation of file organization techniques: single level, two-level and hierarchical level
15	Implementation of File Allocation concept using the technique indexed and link list, Implementation of Contiguous File Allocation using Compaction algorithm

CSC 3109 Software Engineering Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Software Engineering	Semester: 6 Credits: 3
Course Code: CSC 3109	Course Type: Core
Course Area/Domain: Computer Science	Contact hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

COURSE OBJECTIVES

This course has been designed is to provide the knowledge of basic SW engineering methods and practices, and their appropriate application and general understanding of software process models such as the waterfall and evolutionary models. Students will also get an understanding of software requirements and the SRS document and different software architectural styles. This course will also provide an understanding of approaches to verification and validation including static analysis, and reviews software testing approaches such as unit testing and integration testing.

COURSE DESCRIPTION

Nature of Software, Overview of Software Engineering, Professional software development, Software engineering practice, Software process structure, Software process models, Agile software Development, Agile process models, Agile development techniques, Requirements engineering process, Functional and non-functional requirements, Context models, Interaction models, Structural models, behavioral models, model driven engineering, Architectural design, Design and implementation, UML diagrams, Design patterns, Software testing and quality assurance, Software evolution, Project management and project planning, configuration management, Software Process improvement.

TEXT/REFERENCE BOOKS

Software Engineering, A Practitioner's Approach, Pressman R. S.& Maxim B. R., McGraw-Hill. Software Engineering, Sommerville I., Pearson Inc.

CLO	Description	Domain	BT Level*
1	Explain software process models, the software development lifecycle, and the importance of systematic engineering approaches to software development	C	1
2	Analyze user needs and system requirements to create software requirement specifications (SRS) and design system models.	C	3
3	Apply software design principles, architectural patterns, and testing techniques to develop a well-structured, reliable software design.	C	4
4	Function effectively as a team member to manage a software project, demonstrating professional practice and communication through project artifacts.	C	2

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1			X							
CLO.2		X								
CLO.3					X					
CLO.4			X							

GA	Description
2.	Ability to identify, formulate, research literature, and analyze complex problems reaching substantiated conclusions using basic principles of mathematics, natural sciences and computer science.
3	Ability to design solutions for complex problems and design software systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
5	Ability to create, select and apply appropriate techniques, resources, and modern IT tools, including prediction and modeling, to complex activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Nature of Software, Overview of Software Engineering, Professional software development, Software engineering practice.
2	Software process structure, Software process models, Process activities, Software Process improvement.
3	Agile software Development, Agile process models, Agile development techniques.
4	Requirements engineering process, Functional and non-functional requirements.
5	System Modeling: Interaction models, Structural models.
6	System Modeling: Behavioral models, Context models
7	UML diagrams
8	Midterm Exam
9	Architectural design, Design and implementation, Model Driven Engineering
10	Software Design Patterns.
11	Software Testing and Quality assurance.
12	Software evolution.
13	Project management.
14	Project planning.
15	Configuration management.

CSC 3108 Advance Database Management Systems (2,0)

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Advanced Database Management Systems	Semester: 5 Credits: 2
Course Code: CSC 3108	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code(If any): CSCL 3108	Lab Course Name: Lab: Advanced Database Management Systems
Course Pre-Requisite:	

CLO	Description	Domain	BT Level*
1	Apply advanced SQL features including stored procedures, functions, triggers, and cursors for complex database operations.	C	3
2	Analyze transaction processing, concurrency control mechanisms, and database recovery techniques in multi-user environments.	C	4
3	Evaluate distributed database architectures, replication strategies, and data warehousing principles for enterprise applications.	C	5
4	Design and implement solutions using emerging database technologies including NoSQL systems and big data platforms.	C	6

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GA)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				X						
CLO.4					X					

GAs	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Introduction to Advanced DBMS Concepts – Overview, evolution, and scope of advanced databases.
2	Review of SQL and Relational Database Concepts – Normalization, relational algebra, constraints, and advanced querying basics.
3	Advanced SQL: Stored Procedures – Syntax, parameters, control structures, and use cases.
4	Advanced SQL: Functions and Triggers – Creating functions, triggers for enforcing business rules, examples, and best practices.
5	Advanced SQL: Cursors and Dynamic SQL – Cursor types, fetching, iteration, and dynamic query execution.
6	Transaction Management – ACID properties, transaction states, schedules, and serializability.
7	Concurrency Control Mechanisms – Lock-based protocols, timestamp ordering, optimistic concurrency control, and deadlock handling.
8	Recovery Techniques – Checkpointing, log-based recovery, deferred and immediate update strategies.
9	Distributed Database Concepts – Overview, architectures, fragmentation, replication, and allocation strategies.
10	Distributed Query Processing – Query decomposition, optimization, and execution in distributed environments.
11	Replication and Consistency Models – Synchronous/asynchronous replication, eventual consistency, and conflict resolution.
12	Data Warehousing Concepts – OLAP vs OLTP, ETL processes, star and snowflake schemas, and data marts.
13	Emerging Database Technologies – NoSQL systems: key-value, document, column-family, graph databases.
14	Big Data Platforms – Hadoop ecosystem, Spark SQL, distributed storage, and analytics frameworks.
15	Case Studies, Project Presentations, and Revision – Real-world applications, integrating advanced SQL, distributed DB, and big data solutions.

CSCL 3108 Lab: Advance Database Management Systems (0,1)

Course Information Sheet

Programme: Computer Science	Degree: BS
Course: Lab: Advanced Database Management Systems	Semester: 5 Credit: 1
Course Code: CSCL 3108	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Theory Course Code: CSC 3108	Theory Course Name: Advanced Database Management Systems
Course Pre-Requisite:	

CLO	Description	Domain	BT Level*
1	Implement advanced database programming using stored procedures, functions, and triggers in a commercial DBMS.	C	3
2	Demonstrate transaction management, concurrency control, and recovery techniques in multi-user database environments.	C	3
3	Design and implement distributed database operations a data warehousing ETL processes.	C	6
4	Apply NoSQL databases and big data tools to solve data management problems beyond traditional relational moc	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GA)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				X						
CLO.4					X					

GAs	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

CSC 1205 Technical and Business Writing Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Technical and Business Writing	Semester: 6 Credits: 3
Course Code: CSC 1205	Course Type: Core
Course Area/Domain: GER	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 1102	

COURSE OBJECTIVES

The course aims at developing technical and business writing skills of BS Computer science students. Technical Writing prepares students to design effective technical documents for both written and digital media, with particular emphasis upon technical memos, problem-solving and decision-making reports, and organizational, product-support, and technical-information webs.

COURSE DESCRIPTION

Technical and Business writing is a fundamental course for virtually anyone working in the field of science, engineering or business. It is a blend of theory and practice of varied technical and business writing basics, techniques and applications students are likely to encounter in their academic courses and their future careers.

TEXT/REFERENCE BOOKS

Technical Report Writing Today, Pauley Riordan, 8th Edition.
Effective Technical Communication by Ashraf Rizwi, Tata McGraw-Hill.

CLO	Description	Domain	BT Level*
1	Identify the purpose, audience, and context of various technical and business documents	C	1
2	Analyze professional documents to evaluate their effectiveness in terms of structure, tone, and audience suitability.	C	4
3	Create professional-quality documents (letters, Memorandums, Emails, Text Messages, Business Proposals, Reports, etc.) using appropriate formatting, visual aids and technical language.	C	6
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1							X			
CLO.2							X			
CLO.3									X	

GA	Description
7.	Communicate effectively with the computing community about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
9	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

SYLLABUS

Week	Topics
1	Course Overview, what is technical and business writing? Attributes and benefits of technical and business writing
2	What is technical and business writing? Attributes and benefits of technical and business writing
3	Technical narration, description and argumentation, persuasive strategy
4	Brain storming, organization of information, construction of a formal outline.
5	Technical Definitions, Analyzing Different manuals for product descriptions
6	Analyzing different manuals for processes and procedures
7	Memorandums in Technical and Business writing
8	Mid-Term
9	Emails and Text Messages in Technical and Business writing
10	Writing Business/Project proposals
11	Writing Research Proposals
12	Writing short/long reports/types of reports
13	Abstracts and summaries in Technical Writing
14	Preparing and Delivering Effective Oral presentations
15	Individual Presentations

CSC 3110 Community Service

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Technical and Business Writing	Semester: VI Credits: 2
Course Code: CSC 3110	Course Type: Core
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Objectives

Develop an understanding of community service, social responsibility, and ethical engagement.
Apply technical and problem-solving skills to contribute to community-based initiatives.
Enhance teamwork, leadership, and communication through hands-on social projects.
Foster awareness of societal challenges and strategies for impactful change.
Encourage lifelong civic engagement and volunteerism.

Course Description

This course provides students with experiential learning opportunities through community engagement, volunteerism, and social impact projects. Students will collaborate with NGOs, community organizations, and social enterprises to address real-world challenges, applying their technical skills and AI knowledge to support community development. The course fosters civic responsibility, teamwork, and ethical decision-making.

Text/Reference Books

David Bornstein – How to Change the World: Social Entrepreneurs and the Power of New Ideas (Oxford University Press, 2010)
Robert Putnam – Bowling Alone: The Collapse and Revival of American Community (Simon & Schuster, 2000)
Peter Singer – The Life You Can Save: Acting Now to End World Poverty (Random House, 2019)

SYLLABUS:

Week	Topics
1	Introduction to Community Service: Importance, Scope, and Benefits
2	Social Responsibility and Ethical Engagement in AI
3	Identifying Community Needs: Research and Analysis
4	Volunteerism and Non-Profit Organizations: Roles and Responsibilities
5	Project Planning: Defining Objectives and Setting Goals
6	Community Outreach: Working with NGOs and Social Initiatives
7	Fundraising, Grant Writing, and Resource Allocation
8	Midterm Review & Progress Presentation
9	Implementing Service Projects: Hands-on Community Work
10	AI for Social Good: Leveraging Technology for Community Development
11	Leadership and Teamwork in Social Impact Projects
12	Evaluating Social Projects: Measuring Impact and Outcomes
13	Presentation Skills for Social Advocacy
14	Reflection and Lessons Learned from Community Engagement
15	Final Project Submission: Case Studies & Recommendations

Course Learning Outcomes (CLOs)

CLO	Description	Domain	BT Level
1	Understand the importance of community service and social impact.	C	2
2	Identify and analyze societal challenges using research-based approaches.	C	3
3	Apply teamwork, leadership, and communication in a service-learning environment.	A	4
4	Demonstrate ethical decision-making and responsible AI use for social good.	A	5

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	GRADUATE ATTRIBUTES (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1								X		
CLO.2			X							
CLO.3						X				
CLO.4									X	

GRADUATE ATTRIBUTES (GAs)

GAs	Description
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
6	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.
8	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
9	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

CSC 3209 Computer Networks and Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Computer Network and Data Communications	Semester: 4 Credits: 2
Course Code: CSC 3209	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil

COURSE OBJECTIVES

By the end of this course, students will be able to:

1. Build an understanding of the fundamental concepts of computer networking.
2. Familiarize the student with the basic taxonomy and terminology of the computer networking area.
3. Introduce the student to advanced networking concepts, preparing the students for entry Advanced courses in computer networking.
4. Allow the student to gain expertise in some specific areas of networking such as the design and maintenance of individual networks.

COURSE DESCRIPTION

This course provides the students with an overview of the concepts and fundamentals of data communication and computer networks. Topics to be covered include : data communication concepts and techniques in a layered network architecture, communications switching and routing, types of communication, network congestion, network topologies, network configuration and management, network model components, layered network models(OSI reference model, TCP/IP networking architecture) and their protocols, various types of networks (LAN,MAN,WAN and Wireless networks) and their protocols.

TEXT/REFERENCE BOOKS

Computer Networking: A Top-Down Approach Featuring the Internet, 6th edition by James F. Kurose and Keith W. Ross, James F. Kurose and Keith W. Ross , null Edition
Data and Computer Communications, 10th Edition by William Stallings, William Stallings, 1st Edition, 2018.
Data Communication and Computer Networks, 5th Edition by Behrouz A. Forouzan, Behrouz A. Forouzan

CLO	Description	Domain	BT Level*
1	Describe the key terminologies and technologies of computer networks	B	1
2	Explain the services and functions provided by each layer in the Internet protocol stack.	C	2
3	Analyze working and performance of key technologies, algorithms and protocols	C	4
4	Build Computer Network on various Topologies	C	6
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		x								
CLO.2			x							
CLO.3				X						
CLO.4					X					

GA	Description
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Introduction to communication and networks, internet, network edge. Importance of protocols and algorithms to provide seamless communication. Necessity of layered architecture for globalization in communication networks.
2	OSI/ TCPIP layers, Network core, circuit switching and packet switching network performance metrics : delay, path loss, throughput
3	Network types, LAN MAN WAN, latest networks i.e. low power WAN. Future of networks- IoT. communication medium and types of medium (wired wireless)
4	Application layer: principles and protocols http, FTP, SMTP, DNS, peer-to-peer applications. Socket programming concepts.
5	Transport layer, Multiplexing and de multiplexing transport layer protocols TCP, UDP (connection less and connection oriented protocols).
6	The network layer: devices of network layer. Internet protocol, routing in internet
7	Network Layer (Layer 3) wireless protocols for adhoc and sensor networks, Addressing, Sub netting
8	Mid Term Paper.
9	Introduction to link layer, MAC and LLC, Medium access techniques, link virtualization, data center networking.
10	Medium access techniques ALOHA, Slotted Aloha, CSMA/CD, CSMA/ CA
11	Medium Access Control protocols- LLC protocols ARP, RARP
12	Wireless and mobile networks, an introduction of IEEE 802.11, cellular network, mobile IP, IEEE 802.15.4 and IEEE 802.15.6
13	Multimedia networking, streaming and stored video, Voice over IP
14	Network security fundamentals, cryptography
15	Network management. Infrastructure and internet standard management framework. Ubiquitous networks- combination of AI and networks.

CSCL 3209 Lab Computer Networks

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Computer Networks and Data Communications	Semester: 4 Credits: 1
Course Code: CSCL 3209	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Objectives

The objective of this lab session would be to give students about two popular tools used in networking for monitoring and building different types of networks and topologies provided by a well-known protocol/ standard under different vendors

Course Description

Goal of this course is to give some insight into the rationale of why networks are structured the way they are today and to understand the issues facing the designers of next-generation data networks. Much of the course focuses on network algorithms and their performance. Students are expected to have a strong mathematical background and an understanding of probability theory. Topics discussed include: layered network architecture, Link Layer protocols, high-speed packet switching, queuing theory, Local Area Networks, and Wide Area Networking issues, including routing and flow control protocols.

TEXT/REFERENCE BOOKS

Computer Networks by Andrew S. Tanenbaum 5th Edition Pearson
Data Communications and Networking by Behrouz A. Forouzan ,5th Edition , 978-0073376226

CLO	Description	Domain	BT Level*
2	Identify different components and their respective roles in a computer communication system	C	1
3	Apply the knowledge, concepts and terms related to data communication and networking	C	3
4	Solve problems in networking by referring to problems	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3				x						

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Introduction to Wireshark
2	Hyper Text Transfer Protocol(HTTP) using WireShark
3	Domain Name System(DNS) using WireShark
4	Transport Control Protocol (TCP) using WireShark
5	Socket Programming
6	User Datagram protocol(UDP) using WireShark
7	IP(Internet Protocol) using WireShark
8	ICMP(Internet Control Message Protocol) using WireShark
9	DHCP (dynamic host configuration protocol) using WireShark
10	Ethernet and (ARP)Address Resolution protocol using WireShark
11	Configuring VLAN using Packet Tracer
12	802.11 using Wireshark
13	SSL using WireShark
14	NAT(Network Address Translation Protocol) using WireShark
15	Configuring VLAN using Packet Tracer

CSC 3206 Artificial Intelligence Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Artificial Intelligence	Semester: 3 Credits: 2
Course Code: CSC 3206	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week.
Corresponding lab Course code (If any): CSCL 3206	Lab Course Name: Lab Artificial Intelligence
Course Pre-Requisite: CSC 1208, CSCL 1208	

Course Objectives

The objective of the course is to present an overview of artificial intelligence (AI) principles and approaches. Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic, and learning.

Course Description

Artificial Neural Network includes subject of the McCulloch-Pitts Model, activation functions, loss and cost functions, feed-forward and feed-back network structures, Single Layer perceptron's, Multi-Layer perceptron's, Radial-based function networks, weighted networks, perceptron learning algorithms, strategies for avoiding overfitting and underfitting, accelerating convergence and recurrent networks.

TEXT/REFERENCE BOOKS

Artificial Intelligence: A Modern Approach, Russell, S., & Norvig, P., (2013), Prentice Hall. Prolog Programming for Artificial Intelligence, Bratko, I., (2012), Addison-Wesley. Machine Learning, Mitchell, T., (1997), McGraw-Hill

CLO	Description	Domain	BT Level*
1	Understand key components in the field of artificial intelligence	C	2
2	Implement classical artificial intelligence techniques	C	3
3	Analyze artificial intelligence techniques for practical problem solving	C	4
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		x								
CLO.2				x						
CLO.3										x

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Overview of AI Problems; Intelligent Behavior: Turing Test, Formal Definitions for Artificial Intelligence, History and Evolution of Artificial Intelligence
2	Intelligent agents and agents environments, Single versus Multi agent.
3	Knowledge based systems, Nature of Agents: Autonomous versus Semi-autonomous, Reflexive, Goal-based and Utility-based.
4	Tree and Graph, Search Strategies, DFS, BFS, Heuristically Informed Searches,
5	Hill Climbing, A* Search. Adversarial Search, time and space complexity
6	Problem Solving, Problem Representation, Components of Problem Solving, The Two– One Problem.
7	Propositional and Predicate Logic. Pros and Cons of propositional logic, First Order Logic.
8	Midterm Exam
9	Syntax and Semantics of FOL, Using FOL, Atomic/Complex Sentences, Rules of Inference.
10	Proof, Resolution Inference Rule, Resolution Proof.
11	Forward chaining algorithm, backward chaining algorithm.
12	Classification, Inductive Learning, Simple Statistical Based Learning,
13	Machine Learning: Decision Tree
14	Neural Networks, Artificial Neural Networks, Single Layer Feed Forward neural net.
15	Multilayer feed forward neural networks, multilayer back propagation neural networks

CSCL 3206 Lab Artificial Intelligence

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Lab Artificial Intelligence	Semester: 3 Credits: 1
Course Code: CSCL 3206	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 1208, CSCL 1208	

Course Objectives

The objective of the course is to present an overview of artificial intelligence (AI) principles and approaches. Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic, and learning.

Course Description

This course gives a broad overview of the fundamental theories and techniques of Artificial Intelligence. Major topics covered in the lectures include: Overview of AI Problems; Intelligent Behavior: Turing Test, Rationale versus Non-rationale Reasoning; Problem Characteristics: Fully versus Partially Observable, Single versus Multi agent; Intelligent Agents: reactive, deliberative, goal-driven, utility-driven, and learning agents; Uninformed Search: Depth First, Breadth First, Depth First with Iterative Deepening; Informed Search: Hill climbing, A*- Search and their Time and Space Complexity, Local Search, Genetic Algorithm; Game Playing: Min-max, Evaluation functions, Alpha-beta pruning; Propositional and Predicate Logic; Resolution and Theorem Proving; Forward and Backward Chaining; Machine Learning: Introduction, Supervised learning: Inductive learning, Decision tree, Artificial neural networks.

TEXT/REFERENCE BOOKS

Artificial Intelligence: A Modern Approach, Russell, S., & Norvig, P., (2013), Prentice Hall.
 Prolog Programming for Artificial Intelligence, Bratko, I., (2012), Addison-Wesley.
 Machine Learning, Mitchell, T., (1997), McGraw-Hill

CLO	Description	Domain	BT Level*
1	Understand key components in the field of artificial intelligence	C	2
2	Implement classical artificial intelligence techniques	C	3
3	Analyze artificial intelligence techniques for practical problem solving	C	4

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2			X							
CLO.3										X

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Python Introduction – Data Structures and Loops
2	Python Introduction – Functions, Tuples, Sets and Dictionaries
3	Searching – Uninformed searches -Breadth First Search
4	Uninformed searches -Depth First Search
5	Heuristic Search – Greedy Best search, A* Search
6	Genetic Algorithms
7	Prolog Introduction
8	Complex Knowledgebase in Prolog
9	Expert Systems in Prolog, FWD BWD
10	Introduction to Matlab
11	Linear Regression
12	Linear Classification – Decision Tree
13	Regression and Classification using K nearest Neighbors
14	Classification using Single layer Artificial Neural Networks
15	Classification using Multi-layer Artificial Neural Networks

CSC 4106 Parallel and Distributed Computing

Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Parallel and Distributed Computing	Semester: 8 Credits: 3
Course Code: CSC 4106	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: CSC 3107, CSCL 3107	

Course Description

Asynchronous/synchronous computation/communication, concurrency control, fault tolerance, GPU architecture and programming, heterogeneity, interconnection topologies, load balancing, memory consistency model, memory hierarchies, Message passing interface (MPI), MIMD/SIMD, multithreaded programming, parallel algorithms & architectures, parallel I/O, performance analysis and tuning, power, programming models (data parallel, task parallel, process-centric, shared/distributed memory), scalability and performance studies, scheduling, storage systems, synchronization, and tools (Cuda, Swift, Globus, Condor, Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, Open MP, Hadoop, FUSE).

Course Objectives

The objective of this course is to introduce the fundamentals of parallel and distributed processing, including system architecture, programming model, and performance analysis. It will focus on the basic architectural, programming, and algorithmic concepts in the design and implementation of parallel and distributed applications.

TEXT/REFERENCE BOOKS

1. Distributed Systems: Principles and Paradigms, A. S. Tanenbaum and M. V. Steen, Prentice Hall, 2nd Edition, 2007
2. Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet, K Hwang, J Dongarra and GC. C. Fox, Elsevier, 1st Ed.

CLO	Description	Domain	Bt level*
1	Explain and differentiate parallel and distributed computing models and their architectures.	C	2,4
2	Design and implement parallel programs and algorithms using OpenMP and MPI.	C	3,6
3	Analyze and evaluate the performance and bottlenecks of parallel programs	C	4,5
4	Explain and apply distributed system concepts including clock synchronization, mutual exclusion, and election algorithms.	C	2,3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2		X			X					
CLO.3			X		X					
CLO.4		X	X							

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5.	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

SYLLABUS

Week	Topics
1	Asynchronous/synchronous computation/communication.
2	Concurrency control, fault tolerance
3	GPU architecture and programming, heterogeneity, interconnection topologies,
4	Load balancing, memory consistency model, memory hierarchies,
5	Message passing interface (MPI), MIMD/SIMD,
6	Multithreaded programming, parallel algorithms & architectures
7	Parallel I/O, performance analysis and tuning
8	Midterm Exam
9	Power, programming models data parallel, task parallel,
10	Process-centric, shared/distributed memory, scalability and performance studies
11	Scheduling, storage systems, synchronization
12	Tools (Cuda, Swift, Globus, Condor
13	Amazon AWS, OpenStack
14	Cilk, gdb, threads
15	MPICH, Open MP, Hadoop, FUSE).

CSC 4109 Professional Practices Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Professional Practices	Semester: 7 Credits: 2
Course Code: CSC 4109	Course Type: Core
Course Area/Domain: General	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: Nil	

Course Description

This course introduces students to the professional, ethical, legal, and social responsibilities of computing professionals. Topics include computing ethics, organizational structures, software liability, intellectual property rights, employee relations, HR management in IT, health and safety, professional codes of conduct (BCS, IEEE, ACM), accountability, auditing, and social applications of ethics. The course emphasizes real-world case studies and promotes critical thinking to prepare students for responsible professional practice.

Course Objectives

By the end of the course, students will be able to:

1. Understand the foundations of computing professionalism, ethics, and legal responsibilities.
2. Analyze organizational structures, HR practices, software contracts, and intellectual property issues.
3. Apply professional codes of conduct and ethical frameworks (BCS, IEEE, ACM) to real-world scenarios.
4. Evaluate social, legal, and professional responsibilities of computing professionals, including auditing and accountability.

TEXT/REFERENCE BOOKS

Professional Issues in Software Engineering by Frank Bott, Allison Coleman, Jack Eaton and Diane Rowland, CRC Press; 3rd Edition

Computer Ethics by Deborah G. Johnson, Pearson; 4th Edition

ACM/IEEE – Software Engineering Code of Ethics and Professional Practice

CLO	Description	Domain	BT Level*
1	Explain fundamental concepts of computing professionalism, ethics, and philosophy of ethics.	C	2
2	Analyze organizational structures, employee relations, HR practices, and software contracts in the context of professional ethics.	C	4
3	Apply professional codes of conduct (ACM, IEEE, BCS) and ethical frameworks to real-world scenarios.	C	3
4	Evaluate legal responsibilities, accountability, auditing, and social implications of computing practices.	C	5
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1								X		
CLO.2			X							
CLO.3									X	
CLO.4								X		

GA	Description
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
8	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
9	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

SYLLABUS

Week	Topics
1	Computing Profession, Computing Ethics, Philosophy of Ethics.
2	The Structure of Organizations and its effects on Professional ethics
3	Finance and Accounting
4	Anatomy of a Software House, Computer Contracts
5	Intellectual Property Rights
6	The Framework of Employee Relations Law and Changing Management Practices
7	Human Resource Management and IT
8	Midterm Exam
9	Health and Safety at Work
10	Software Liability, Liability and Practice
11	Computer Misuse and the Criminal Law, Regulation and Control of Personal Information
12	Overview of the British Computer Society Code of Conduct, IEEE Code of Ethics
13	ACM Code of Ethics and Professional Conduct, ACM/IEEE Software Engineering Code of Ethics and Professional Practice
14	Accountability and Auditing
15	Social Application of Ethics.

CSC 4107 Information Security Course Information Sheet

Programme: CS/SE/AI	Degree: BS
COURSE: Information Security	Semester: 7 Credits: 2
Course Code: CSC 4201	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): CSCL 4107	Lab Course Name: Lab: Information Security
Course Pre-Requisite: Nil	

Course Description

This course provides a comprehensive foundation in Information Assurance and Information Security. It covers core security principles, international standards, cyber security threats, preventive measures, cryptographic systems, authentication mechanisms, and network security fundamentals. Students learn to evaluate risks, analyze attacks, implement passive and active defense mechanisms, and examine security policies. The course emphasizes hands-on understanding through case studies, focusing on designing and auditing secure information systems according to industry standards.

Course Objectives

By the end of the course, students will be able to:

1. Understand fundamental principles, standards, and terminology of Information Assurance and Information Security.
2. Identify cybersecurity threats and apply suitable preventive, cryptographic, authentication, and policy-based security measures.
3. Analyze basic network concepts, typical network attacks, and implement passive and active network security mechanisms.
4. Design, evaluate, and audit security systems using real-world case studies and enterprise-level security practices.

TEXT BOOKS

Stallings & Brown, Computer Security, Principles and Practice

Whitman & Mattord, *Principles of Information Security*

Behrouze A Forouzan, Cryptography and Network Security, McGraw Hill Publishers

CLO	Description	Domain	BT Level*
1	Explain principles, standards, concepts, and trade-offs in Information Assurance and Information Security.	C	2
2	Identify threats and apply preventive, cryptographic, authentication, and policy-based security measures.	C	1,3
3	Analyze network concepts, attacks, and implement passive & active measures to design and audit secure networks.	C	4
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2		X								
CLO.3			X							

GA	Description
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

SYLLABUS

Week	Topics
1	Introduction to Information Assurance Information Security Principles
2	Standards of Information Assurance
3	Terms, Concepts and Tradeoffs Related to Information Security
4	Introduction to Cyber Security Cyber Security Threats and Measures
5	Cyber Security SOPs and Policies
6	Introduction to Cryptography
7	Symmetric Cryptography
8	Midterm Exam
9	Asymmetric Cryptography
10	Secure Identity, Authentication and Signature Techniques
11	Understanding Basic Network Concepts
12	Typical Network Attacks and Threats from the Internet
13	Passive Measures to Ward off Network Attacks
14	Active Measures to Safeguard an Enterprise Network
15	Case Study for Designing and Auditing a Security System

CSCL 4107 Lab: Information Security

Course Information Sheet

Programme: Computer Science	Degree: BS
Course: Lab Information Security	Semester: 7 Credits: 1
Course Code: CSCL 4107	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Course Code (If any): CSC 4107	Course Name: Information Security
Course Pre-Requisite: None	

Course Description

The lab course provides hands-on experience in Information Security techniques, including cryptography, network security, threat analysis, authentication mechanisms, and auditing security systems. Students will implement concepts learned in theory, analyze security threats, and evaluate enterprise security measures through practical exercises.

Course Objectives

By the end of this course, students will be able to:

1. Demonstrate practical understanding of information security principles, standards, and policies.
2. Apply cryptographic techniques, authentication, and identity management in hands-on scenarios.
3. Analyze network threats and implement passive and active security measures.
4. Design, evaluate, and audit secure systems through real-world case studies.

Text/Reference Books

Stallings & Brown, Computer Security, Principles and Practice

Behrouze A Forouzan, Cryptography and Network Security, McGraw Hill Publishers

Lab manuals, online tutorials for Wireshark, OpenSSL, Kali Linux

CLO	Description	Domain	BT Level*
1	Demonstrate fundamental information security concepts and standards using practical exercises..	C	2
2	Apply cryptographic techniques, identity management, and authentication mechanisms in practical scenarios.	C	3
3	Implement network security measures and perform threat analysis on networks and systems..	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affectivedomain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2		X								
CLO.3			X							

GA	Description
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

SYLLABUS

Week	Topics
1	Introduction to IS Lab & Tools Setup
2	Information Assurance Principles Exercise
3	Standards & Policies Simulation
4	Cyber Threats Identification
5	Symmetric Cryptography Lab
6	Asymmetric Cryptography Lab
7	Hashing & Digital Signatures Lab
8	Authentication Techniques Lab
9	Network Concepts Lab
10	Network Attacks Simulation
11	Passive Security Measures Lab
12	Active Security Measures Lab
13	Vulnerability Analysis Lab
14	Case Study – Designing Secure System
15	Audit & Evaluation Lab

CSC 4105 Final Year Project-I

Course Information Sheet

Programme: Computer Science	Degree: BS
Course: Final Year Project-I	Semester: 7 Credits: 3
Course Code: CSC 4105	Course Type: Core
Course Area/Domain: Computer Science	Contact Hours: 3 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Description

This course provides students with the opportunity to identify, design, and plan a computing project under faculty supervision. Emphasis is on project proposal development, requirements gathering, system design, literature review, feasibility analysis, and preparation for implementation. Students learn to apply software engineering principles, document their work professionally, and communicate project objectives effectively.

Course Objectives

By the end of the course, students will be able to:

1. Identify and define a computing problem suitable for a final-year project.
2. Perform literature review, feasibility analysis, and requirement specifications for the project.
3. Design a solution architecture and develop project documentation, including project proposals and design documents.
4. Demonstrate professional communication, teamwork, and project planning skills.

Text/Reference Books

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CLO	Description	Domain	BT Level*
1	Identify and define a complex computing problem and its objectives.	C	1
2	Conduct literature review, requirement gathering, and feasibility analysis for the proposed project.	C	4
3	Design a project solution architecture and prepare professional project documentation.	C	6
4	Demonstrate teamwork, communication, and project management skills during project planning and proposal presentation	A	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1			X							
CLO.2		X								
CLO.3				X						
CLO.4						X	X			

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
6	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.
7	Communicate effectively with the computing community about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

SYLLABUS

Week	Topics
1	Course introduction, project requirements, selection of project domain
2	Problem identification and scope definition
3	Literature review methodology, research sources
4	Literature review – submission of proposal
5	Requirement gathering techniques and documentation
6	Feasibility study: technical, economic, and operational feasibility
7	UML diagrams: use-case, class, sequence diagrams
8	Solution design: architecture, modules, data flow
9	Midterm review: project proposal draft discussion
10	Prototype planning / mock-ups
11	Documentation: project plan, Gantt chart, and design report
12	Teamwork coordination and peer review sessions
13	Finalizing project 1 st part and design documentation
14	Project presentation rehearsal and feedback
15	Final submission of FYP-1 & presentation

AIC 2401 Programming for Artificial Intelligence

Course Information Sheet

Programme: CS/SE/AI	Degree: BSAI
Course: Programming for Artificial Intelligence	Semester 4 Credits: 2
Course Code: AIC 2401	Course Type: AI Core
Course Area/Domain: Artificial Intelligence	Contact Hours: (2,1)
Corresponding Lab Course Code (If any): AICL 2401	Lab Course Name: Lab Programming for Artificial Intelligence
Course Pre-Requisite: CSC 3206, CSCL 3206	

COURSE OBJECTIVES

The objectives of this course are to provide comprehensive and in-depth knowledge of AI principles and techniques by introducing AI's fundamental problems, and the state-of-the-art models and algorithms used to undertake these problems. This course is also designed to expose students to the frontiers of AI-intensive computing and information systems, while providing a sufficiently strong foundation to encourage further research.

COURSE DESCRIPTION

This course covers the basic concepts of various fields of artificial intelligence like Artificial Neural Networks, Natural Language Processing, Machine Learning, Deep Learning, Genetic algorithms etc., and its implementation perspectives.

TEXT/REFERENCE BOOKS

1. Artificial Intelligence with Python, Prateek Joshi, Packt Publishers, 2nd Edition, 2020.
2. Introduction to Logic Programming, Michael Genesereth, Vinay K Chaudhri, et al., Morgan & Claypool, February 2020.

CLO	Description	Domain	BT Level*
1	Develop basic understanding of the building blocks of AI	C	1
2	Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models	C	2
3	Demonstrate proficiency in applying scientific method to models of machine learning.	C	3

*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2				x						
CLO.3										x

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Introduction to the Artificial Intelligence, Introduction to programming languages for Artificial Intelligence.
2	Basics of Python, Data Types and Operators, control flow, Functions, Scripting and Classes.
3	Introduction to Data Processing, Data Extraction, Data Labeling, Data Formatting
4	Machine Learning Concepts, Modeling, Over and under fitting, Correctness, Bias vs. variance, Feature extraction
5	Introduction to Un-Supervised learning, Supervised learning, Reinforcement learning
6	Clustering, Partition Clustering, kNN – Clustering, Hierarchical Clustering, Graph Clustering
7	Association rules mining, Positive and negative association rule mining
8	Mid Term Paper.
9	Linear Regression, Gradient Descent, Maximum Likelihood estimation
10	Supervised learning techniques, K Nearest Neighbors classification model, Dimensionality
11	Decision Trees, ID3, Random Forests
12	Artificial Neural Networks, Types of neural networks, Perceptron, Feed forward Neural Networks, Back Propagation
13	Recommender Systems, Recommending on popularity, Collaborative filtering
14	Performance Metrics, Need for Performance Metrics, Key Methods of Performance Metrics, Confusion Matrix Example
15	Natural Language Processing: Critical Entities in NLP, Stemming, Part of Speech Tagging, Lemmatization

AICL 2401 LAB Programming for Artificial Intelligence

Course Information Sheet

Programme: CS/SE/AI	Degree: BSAI
Course: Lab Programming for Artificial Intelligence	Semester 4 Credits:1
Course Code: AICL 2401	Course Type: AI Core
Course Area/Domain: Artificial Intelligence	Contact Hours: 3
Course Pre-Requisite: CSC 3206, CSCL 3206	

COURSE OBJECTIVES

The objectives of this course are to provide comprehensive and in-depth knowledge of AI principles and techniques by introducing AI's fundamental problems, and the state-of-the-art models and algorithms used to undertake these problems. This course is also designed to expose students to the frontiers of AI-intensive computing and information systems, while providing a sufficiently strong foundation to encourage further research.

COURSE DESCRIPTION

This course covers the basic concepts of various fields of artificial intelligence like Artificial Neural Networks, Natural Language Processing, Machine Learning, Deep Learning, Genetic algorithms etc., and its implementation perspectives.

TEXT/REFERENCE BOOKS

1. Artificial Intelligence with Python, Prateek Joshi, Packt Publishers, 2nd Edition, 2020.
2. Introduction to Logic Programming, Michael Genesereth, Vinay K Chaudhri, et al., Morgan & Claypool, February 2020.

CLO	Description	Domain	BT Level*
1	Develop basic understanding of the building blocks of AI	C	1
2	Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models	C	2
3	Demonstrate proficiency in applying scientific method to models of machine learning.	C	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1		X								
CLO.2				X						
CLO.3										X

GA	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

SYLLABUS

Week	Topics
1	Introduction to Python, Data Types and Operators, control flow, Functions, Scripting and Classes.
2	Anaconda, Jupyter Notebook, NumPy, Pandas, and Matplotlib
3	Dataset preparations, Data preprocessing, Data Labeling, Data Formatting, Features Extractions, Null data handling
4	Machine Learning Libraries with Pandas
5	Bayesian Model implementation
6	kNN Clustering and Hierarchical Clustering Implementation
7	Apriori Algorithm – association rule mining
8	Mid Term Paper.
9	Linear regression algorithm
10	KNN classification algorithm
11	Decision Trees - Random Forests algorithms
12	CNN implementation
13	Collaborative filtering algorithm -Recommender Systems
14	Performance metrics creation for the supervised and unsupervised algorithms – comparative analysis
15	Working on Natural Language Tool Kit (NLTK), Apache Open NLP

CSC 3111 Entrepreneurship Course Information Sheet

Programme: CS/SE/AI	Degree: BS
Course: Entrepreneurship	Semester: V Credits: 2
Course Code: CSC 1101	Course Type: General
Course Area/Domain: GER	Contact Hours: 2 hours/Week.
Corresponding Lab Course Code (If any): Nil	Lab Course Name: Nil
Course Pre-Requisite: None	

Course Objectives

Understand entrepreneurial thinking and business innovation.
 Develop a business model using lean startup principles.
 Analyze market opportunities, funding options, and risk management.
 Learn about AI-driven entrepreneurship and emerging business trends.
 Strengthen leadership, teamwork, and problem-solving skills in business contexts.

Course Description

This course introduces the fundamentals of **entrepreneurship, innovation, and business development**, focusing on **AI-driven startups** and tech-based ventures. Students will explore **entrepreneurial mindset, business model development, funding strategies, and market analysis**, with practical insights into launching and scaling a successful startup.

Text/Reference Books:

1. Steve Blank & Bob Dorf – The Startup Owner’s Manual (K&S Ranch Publishing, 2012)
2. Eric Ries – The Lean Startup (Crown Business, 2011)
3. Alexander Osterwalder & Yves Pigneur – Business Model Generation (Wiley,

Reference Books:

1. Peter Thiel – Zero to One: Notes on Startups, or How to Build the Future (Crown Business, 2014)
2. Guy Kawasaki – The Art of the Start 2.0 (Portfolio, 2015)
3. Thomas H. Byers, Richard C. Dorf, Andrew Nelson – Technology Ventures: From Idea to Enterprise (McGraw-Hill, 2018)

SYLLABUS:

Week	Topics
1	Introduction to Entrepreneurship: Concepts, Types, and Role in Economy
2	Entrepreneurial Mindset: Creativity, Risk-Taking, and Problem-Solving
3	Business Idea Generation: Identifying Opportunities in AI and Technology
4	Business Models: Lean Startup, Business Model Canvas, and Value Proposition
5	Market Research and Feasibility Analysis
6	Financial Planning: Revenue Streams, Cost Structure, and Break-Even Analysis
7	AI-Driven Startups: Trends, Challenges, and Opportunities
8	Midterm Exam
9	Business Pitching and Communication Strategies
10	Funding and Investment: Venture Capital, Crowdfunding, and Angel Investors
11	Legal and Ethical Considerations in Entrepreneurship
12	Digital Marketing and Branding for Startups
13	Scaling and Growth Strategies for Tech-Based Startups
14	Risk Management and Crisis Handling in Startups
15	Case Studies: Successful AI Startups and Lessons Learned

COURSE LEARNING OUTCOMES (CLOs)

CLO	Description	Domain	BT Level*
1	Understand the fundamentals of entrepreneurship, business models, and innovation.	C	2
2	Develop a business idea and create a startup model based on market research.	C	3
3	Analyze financial planning, funding strategies, and risk management for startups.	C	4
4	Apply leadership, teamwork, and problem-solving skills in entrepreneurial ventures.	A	5

MAPPING COURSE LEARNING OUTCOMES (CLOs) AND GRADUATE ATTRIBUTES (GAs)

	GRADUATE ATTRIBUTES (GAs)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2				X						
CLO.3					X					
CLO.4						X				

GRADUATE ATTRIBUTES (GAs)

Gas	DESCRIPTION
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.
5	Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6	Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.

CSC 3115 Understanding of Holy Quran-I

Course Information Sheet

Programme: Computer Science	Degree: BS
Course: Understanding of Holy Quran-I	Semester: 5 Credits: 1
Course Code: CSC 3115	Course Type: University Requirement
Course Area/Domain: Islamic Studies	Contact Hours: 1 hour/Week.
Corresponding Lab Course Code(If any): N/A	Lab Course Name: N/A
Course Pre-Requisite: None	

CLO	Description	Domain	BT Level*
1	Demonstrate an understanding of the fundamental teachings of the Holy Quran.	C	1
2	Analyze ethical, moral, and social concepts from the Quran.	C	2
3	Apply Quranic teachings to real-world ethical and social issues.	C	3
4	Develop a sense of responsibility based on Quranic principles.	A	3
*BT=Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A=Affective domain			

	Graduate Attributes (GA)									
	1	2	3	4	5	6	7	8	9	10
CLO.1	X									
CLO.2		X								
CLO.3			X							
CLO.4				X						

GAs	Description
1	Completion of an accredited program of study designed to prepare graduates as computing professionals.
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.

3	Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

SYLLABUS:

Week	Topics
1	Introduction to the Holy Quran – Revelation, structure, compilation, and importance in Islam.
2	Understanding the Quranic Language – Arabic basics, key terms, and interpretation (Tafsir) methods.
3	Themes and Structure of the Quran – Meccan vs. Medinan Surahs, thematic divisions, and chronology.
4	Fundamental Teachings I – Tawheed (Oneness of God) and Prophethood.
5	Fundamental Teachings II – Worship, prayer, and obedience to Allah.
6	Ethical Concepts – Honesty, integrity, justice, and social responsibility
7	Moral Principles – Compassion, patience, forgiveness, and humility.
8	Social Concepts – Family, community relations, equality, and charity.
9	Case Studies: Applying Quranic Ethics – Examples from contemporary social issues.
10	Decision Making Based on Quranic Guidance – Resolving personal and societal dilemmas.
11	Quranic Teachings and Personal Development – Self-discipline, accountability, and character building.
12	Responsibility and Leadership – Quranic principles in governance, work, and social roles.
13	Quranic Stories and Lessons – Prophets, moral parables, and life guidance.
14	Integrating Quranic Knowledge – Reflection exercises and practical applications.
15	Revision, Reflection, and Assessment – Summary, discussion, and presentations on applied learnings.